Breaking Bias: A Context-Aware Multimodal Framework for Detecting and Neutralizing Ideological Bias in News

ABHIJIT MITRA and RAJDEEP MAJUMDAR, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India
RIVUJIT DAS, Department of Computer Science, New York University Courant Institute of Mathematical Sciences, New York, New York, USA
ARGHYADEEP SAHA, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India
JAYANTA PAUL, Department of Computer Application and Science, Institute of Engineering and Management, Kolkata, India
SOUVIK MONDAL and JAYA SIL, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India

Media bias, particularly *bias by word choice*, influences public perception and fosters polarized clusters of opinion. Biased word choices, often subtle and context-independent, erode trust and often jeopardize social frameworks. Detecting and neutralizing such biases remains a challenge, as existing feature-based and neural models fail to capture nuanced ideological leanings, especially in dynamic news contexts. This issue has been addressed by proposing a multimodal framework *CANDOR* (*Context-Aware Analysis for Neutrality and Detection of Objective Reporting*) for detecting and mitigating media bias. CANDOR integrates *ideology-aware text representations* and *context vectors from visual and textual data* in order to improve bias detection and neutralization effectiveness. The framework leverages *BIASLENS*, a novel pre-training objective that contrasts same-story articles of varying ideologies, and *VT-CT*, a vision-text context transformer to enrich contextual embeddings. It detects polar sentences, extracts ideological biases, and generates neutralized content while preserving factual accuracy. CANDOR employs *semantic loss functions* targeting fluency, content preservation, and bias neutralization for natural language generation. Tested on benchmark datasets like *ALLSIDES*, the framework demonstrates a 4–13% *improvement in accuracy* and 1–12% *enhancement in content preservation*, outperforming state-of-the-art models. Human evaluations confirm the framework's ability to neutralize media bias while maintaining readability. This work provides a step toward mitigating media bias and fostering balanced, objective news reporting.

Rivujit Das, Rajdeep Majumdar, Abhijit Mitra, and Arghyadeep Saha contributed equally to this research.

Authors' Contact Information: Abhijit Mitra (corresponding author), Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India; e-mail: abhijitmitra96@gmail.com; Rajdeep Majumdar, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India; e-mail: rajdeepmajumder321@gmail.com; Rivujit Das, Department of Computer Science, New York University Courant Institute of Mathematical Sciences, New York, New York, USA; e-mail: rd3681@nyu.edu; Arghyadeep Saha, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India; e-mail: arghyadeepsaha17@gmail.com; Jayanta Paul, Department of Computer Application and Science, Institute of Engineering and Management, Kolkata, India; e-mail: jayanta.paul@iem.edu.in; Souvik Mondal, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India; e-mail: 2023csb029.souvik@students.iiests.ac.in; Jaya Sil, Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Howrah, India; e-mail: js@cs.iiests.ac.in.

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1 Overview

Subjectivity in natural language pertains to the elements of communication used to convey ideas, assessments, and conjectures, often shaped by an individual's emotional condition and perspectives. Writers and editors frequently endeavor to eliminate biased language; yet, individuals perceive media news as biased [95], and bias constitutes the predominant cause of mistrust in the media [53]. Consequently, while conveying factual information, it is essential to distinguish subjective language from objective language. Bias can manifest through word choice, such as portraying content in a non-neutral manner [163], or via the omission [8] of information. We concentrate on a specific kind of prejudice: *inappropriate subjectivity* ("subjective bias"), which arises when language that ought to be neutral and equitable is distorted by emotion, opinion, or preference, whether intentionally or subconsciously.

Media bias is examined and evaluated in journalism research [44, 67], and in **Natural Language Processing (NLP)** research [35, 90]. Media bias occurs when subjective reporting on a particular incident supplants objective coverage. In the context of social media research, individuals often choose messages, attitudes, or choices that align with their pre-existing views [193]. Table 1 displays news headlines that demonstrate different political opinions, despite three media outlets reporting on the same event. The phrases "9/11 Generation Comes of Age" and "war without end" illustrate the apparent unfavorable stance of both right-leaning media, such as Wall Street Journal, and left-leaning media, like Washington Post, regarding the attack of 9/11. Conversely, the center-leaning biased media outlet, Associated Press, used the phrase "never forget" as a euphemism in memory for the attack on 9/11. Indeed, such polarized signals are prevalent not just in news headlines but also throughout the body content of news stories [91]. The language used in the story "*9/11 will be* '*never forgotten*' *by the United States, even after 18 years.*" from *allsides.com* illustrates media bias at the sentence level (*Supplementary Materials; Section A*, *Table 1*). In order for NLP to identify bias in the sample sentences, it must first extract sentiment keywords; however, it is challenging to ascertain the alignment of these words without previous knowledge of an individual's political opinions or the incident in question. As news delivery becomes more automated, consumers can effortlessly filter out content that contradicts their perspectives. Consequently, it is unsurprising to see increasing polarization in recent years.

To avoid overloading the definitions of "news debias," "news polarity debias(ing)," "news bias mitigation," "bias neutralization," we refer to our kind of text debiasing as *neutralizing bias* of that text. To ensure clarity and precision, we explicitly define "bias neutralization":

> *Neutralization of media bias or polarity debiasing refers to the process of identifying and neutralizing ideological polarity in textual content, thereby eliminating linguistic or contextual cues*

Table 1. Three News Headlines from Washington Post, Associated Press and Wall Street Journal

Media (Polarity)	News Title
Washington Post (L)	"A letter to 9/11: Happy birthday to the *war without end*"
Associated Press (C)	"18 years later, America vows to *'never forget'* 9/11"
Wall Street Journal (R)	"The *9/11 Generation Comes of Age*"

C, center-oriented; L, left-oriented; R, right-oriented. The underlined group of words show how the news headlines demonstrate different political opinions, despite three media outlets reporting on the same event.

that contribute to biased or partisan representations, while preserving the factual and contextual integrity of the original information.

A method to address this issue is to neutralize the polarity of news stories while preserving the substance, making them acceptable to audiences with varying beliefs. Nevertheless, doing this manually is labor-intensive and may create annotator bias. This study presents an automated system capable of converting any articles to its biased neutralized equivalent. Extensive research has been conducted on quantifying subjectivity using text classification algorithms, utilizing linguistic elements to characterize media bias [183, 190], and subsequently refined the pre-trained word embeddings such as BERT [154]. Nevertheless, only a finite number of research endeavors concentrate on the identification and consolidation of bias [34, 115]. The development of automated algorithms for detecting media bias typically relies on the nuanced characteristics of such bias. Consequently, the development of precise quantitative detection methods presents a formidable challenge in media bias research [51, 185, 189, 190]. The current methodologies mostly represent *low-level lexical information*, either explicitly, such as via bag-of-words [60], or implicitly via neural networks [55]. When applied to articles on events that were not part of the training data, these methods do not always work. The rationale is that bias indicators are nuanced and infrequent in papers, particularly those independent of events. Finding article-level bias requires more than just modeling low-level information at the article level. Our objective is to eliminate prejudice in text by proposing modifications that enhance its neutrality. The suggested methodology diverges from previous studies that have mitigated biases in text representations by eliminating prejudicial dimensions from word embeddings [63] and the concealed states of predictive models [43].

In the rest of the article, "debias" or polarity flipping refers to the general changing of polarity by *neutralization* (left-oriented to neural, right-oriented to neutral, and center-oriented to neutral). The framework can accept either headlines or full-length articles as input, as shown in Figure 1. The text is first passed through a polarity detector, which identifies sentences that exhibit polarity and next these sentences are then passed to a polarity flipper. Finally, the output of our framework, that is neutralized text, is merged with the original article. Note that our framework does not change the content of the article. It only changes the way the content is expressed.

Creating such a system has five primary technological obstacles. Political polarization is difficult to discern. In contrast to lexical qualities that explicitly convey attitude or feelings (e.g., "happy," "sad," "angry"), polarity is often inferred from context or extended passages of text. Secondly, self-attention networks have demonstrated encouraging outcomes in numerous NLP tasks [210] wherein the input sequence is regarded as a collection of word tokens, with each token independently attending to the other tokens in the collection. As a result, contextual information is disregarded in these tasks while assessing relationships among the pieces. Numerous researchers [206, 214], have shown that contextual information may enhance how dependencies are modeled in neural

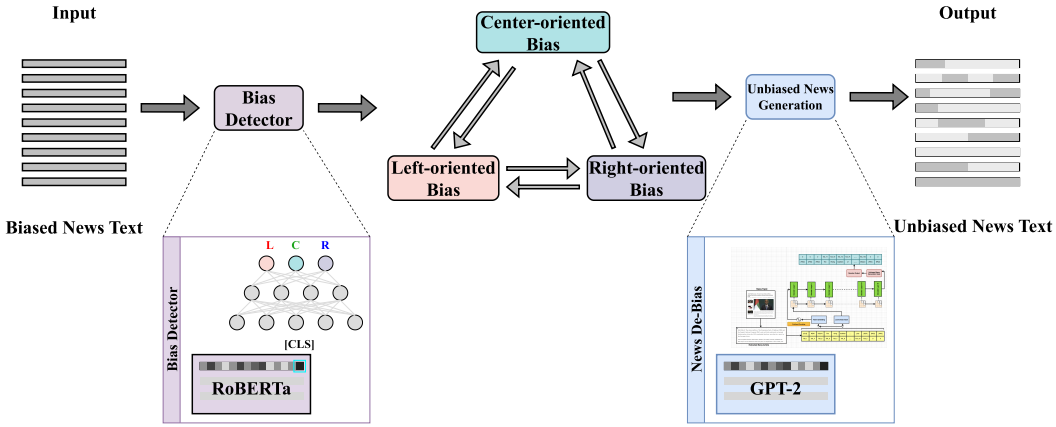


Fig. 1. A synopsis of our framework. The polarity detector will begin by looking for the polar component in the original news headline or text body, which will then be neutralized by the article debiaser or neutralizer.

representations, particularly in attention models. Third, ideology is a pervasive element in political science, social science, journalism, and media studies [134]. Decades of research have been dedicated to quantifying ideology through voting data [110], survey findings [152], social networks [20], campaign contribution records [28], and textual analysis [59]. Each of these methodologies has distinct advantages and disadvantages. While **Pre-Trained Language Models (PLMs)** indicate potential for generalizable solutions, their capacity to assimilate information and identify intricate elements, such as ideology, ideology from texts in a variety of genres is yet unknown. Fourth, even after identifying political polarity and ideology, it remains a challenging endeavor to distinguish neutral information from polar statements in the text. This creates challenges for optimal polarity inversion that just changes the text's polar aspect without changing its substance. The existing benchmark dataset is constrained in both size and variety. Datasets including biased gold standard annotations are currently limited in availability and display several deficiencies.

We tackle the aforementioned difficulties as follows:

—We present a polarity detector that makes use of a fine-tuned language model's attention weights, especially the RoBERTa classifier. We use an efficient domain-adaptive pre-training methodology [138] that depends on a pertinent bias-related dataset. There seems to be a lack of literature about domain-adaptive pre-training as it pertains to media bias. It can identify polar or biased sentences requiring transformation inside an article, enabling us to circumvent unneeded alterations of phrases that do not affect the polarity. Our study's main objective is to determine how domain-adaptive pre-training affects the efficiency of several **Large Language Models (LLMs)** in identifying media bias. Our primary contribution is the usage of transformer-based models while taking biased language into consideration.

—We offer a strategy to enhance the self-attention network by capturing contextual richness while preserving simplicity and flexibility. Hence, *context vectors* are created from the internal representations, so circumventing the need for embeddings of preceding phrases. In particular, we contextualize the shifts between the input, key and query layers, which are used to determine the items' relative importance. We use a variety of contextualization techniques, including: (1) *unified multimodal context* which incorporates the sequence's overall information; (2) *layered multimodal context*, which uses multilayer representations to combine syntactic and semantic information; and (3) *hierarchical fusion context*, which combines the data from the two context vectors given above.

—Ideological bias is often shown via the selection of entities and events for presentation. For example, as noted by Fan et al. [51], one method to get such context is by comparing news pieces on the same topic covered by media with differing viewpoints. We develop a novel PLM, **Bias-aware Ideology and Article Story-driven Learning utilizing Explicit Negative and Same-story (BIASLENS)** contexts. We use continuous pre-training [73], whereby we create an *ideology/philosophy objective* that operates across clusters of *same-story* articles to aggregate articles that have similar ideologies and compare and contrast them with articles that have different ideologies. We add a *story/narrative objective* to further enhance it, that compels the model to concentrate on substantive material rather than excessively depending on shortcuts. Both goals are used in conjunction with our specific mask language model objectives that concentrate on entities and moods.

—Inspired by the recent achievements of LLMs in text generation [123], we use the established Transformer-based encoder–decoder architecture [210] to develop a hybrid multimodal LLM-based **Context-Aware Analysis for Neutrality and Detection of Objective Reporting (CANDOR)** framework [196]. A multimodal model is introduced to include underlying ideology/philosophy of the news, images of the associated news along with the news articles to help the model remove biases from the news articles more accurately. Pre-trained LLM model, BIASLENS and ViT [48] models are fine-tuned on the dataset to generate unbiased news from the biased news. We augment the language model loss with three semantic losses that target the retention of information, the effectiveness of bias neutralization, and the overall fluency of the generated sentences.

—When it comes to pre-training LLMs, one major issue in media bias research is the absence of complete, gold-standard bias datasets. The existing publicly accessible benchmark dataset for political polarity flipping or bias mitigation exhibits limitations, including low **Inter-Annotator Agreement (IAA)**, a small sample size, absence of annotator background information [186, 188], and a lack of source diversity. Moreover, cutting-edge neural language models often need substantial volumes of training data to produce significant representations, which are incongruent with the dimensions of existing media bias datasets. For this job, we used a much bigger (12,265 articles) and more diversified (20 outlets) benchmark dataset, referred to as *ALLSIDES*. The dataset has article polarity labels attached to it and includes other metadata such as themes, article pictures, and impartial equivalents of the news items.

2 Review of Literature

This section analyzes current research on polarity, focusing on polarization, polarity detection, various transformer-based detection methods, domain-adaptive pre-training, context-aware cross-modality attention, ideology prediction, and media bias mitigation or neutralization. At the outset, we compile the findings of previous research on political polarization. Subsequently, we examine several methodologies used in both technological systems and sociological research to ascertain political polarity. Finally, we take a look back at the research on automated style transfer in NLP, showcase the polarity debiasing or bias neutralization challenge as an instance of style transfer in text, and assess the existing state of the art in social science polarity manipulation.

2.1 Political Polarization

The study of political polarization dates back at least to the 1950s and has continued for decades. A multitude of studies indicate that political polarization is present in contemporary media [1, 52]. According to data analysis of newspapers, newsrooms often modify their biases to satisfy potential readers who want a strong alignment between a newspaper's position and their own ideas [56]. Data from cable and broadcast television has yielded a similar conclusion [66]. People continually

prefer congenial information over uncongenial information, according to laboratory experiments, particularly in the context of politics [16]. Longitudinal behavioral research on online political news consumption indicates that selective exposure mostly occurs among those who often engage with partisan news sources [69]. These results underscore the need for a technology that enables the presentation of same knowledge with varying ideological polarity to accommodate individuals with diverse beliefs.

2.2 Polarity Detection

Although various types of media bias exist, such as *bias by omission of information* or *bias by personal perception* [182], our emphasis is on *bias by word choice*, wherein distinct terms denote the same concept [188]. Initial methodologies employed manually designed linguistic features to identify biased news coverage at both the word level [163, 180] and the sentence level [86] using conventional machine learning techniques. Basic lexical data were used in early approaches for bias identification. Recasens et al. [163], for example, assessed whether a Wikipedia statement conveys a neutral viewpoint by using linguistic indicators such as factive verbs and implicatives. The measurement of media bias has been studied by journalism experts in addition to the NLP community. For instance, Gentzkow and Shapiro [56] analyzed the preferences of terms on either side. Their tests demonstrate the terms selected deliberately for political effect. Groseclose and Milyo [67] used think-tank citation counts to assess bias. Scientists in the field of NLP found bias using methods based on neural networks. Iyyer et al. [90] combined the polarity of individual words using RNNs in order to use parse trees to anticipate sentence-level bias. Gangula et al. [55] classified article bias using headline attention. Graph-CNN was used to encode social information by Li and Goldwasser [111].

The examination of media bias identification started with the research conducted by Lin et al. [116]. As of that time, different terminologies have been used to study media bias, including *perspectives* [116], *ideology* [90], *truthfulness* [157], and *hyperpartisanship* [98]. According to Groeling [65], media bias may be divided into two categories: *selective bias* and *presentation bias*. The selection of which events to cover is influenced by selective bias, while presentation bias modifies the way the news is presented. The purpose of this research is to lessen presentation bias. The majority of early research in this field relies on human annotators, with linguistic characteristics proving essential in influencing human assessments of polarity. In their study, Yano et al. [230] collected 1,100 texts that may be biased and asked annotators to identify biased phrases, specifying the political orientation and the corresponding word tokens. Every sentence has five annotations, resulting in a total of 5,205 judgments. Almost fifty percent of the statements are categorized as "*not biased*," whereas merely 2.4% are identified as "*very biased*." Additionally, they provide two lists of the most significant weighted bi-grams—that is, phrases with two or less words—for both the conservative and liberal groups as determined by human annotators.

The aforementioned studies use annotations to manually ascertain political polarity. This may incur significant expenses when managing a substantial volume of data, particularly when the biased material is shown to be rather scarce [24]. As a result, some scholars favor automated methods that use deep neural network capabilities for text analysis. The majority of earlier studies utilized CNN-LSTM [173] or RNN [90] to determine political polarity using syntactic and semantic characteristics. Recent research seeks to implement more sophisticated network architectures owing to the constraints of RNNs in modeling comprehensive news articles [17]. The attention-based multiview model proposed by Kulkarni et al. [104] uses data from the link structure, content, and title of the article to determine the news's ideology. According to Bhatia and Deepak [26] and Menini et al. [135], political ideology can be viewed as a spectrum of sentiment polarities regarding various subjects, another method entails detecting polarity using topical analysis [61]. Our suggested

polarity detector improves qualitative performance by using domain-adaptive learning with an unsupervised language model that has been trained using a massive corpus of data.

2.3 Transformer-Based Polarity Detection Approaches

The transformer architecture [210] has demonstrated exceptional efficacy in numerous downstream tasks [142, 215, 216]. Nevertheless, neural language models like RoBERTa [127] and BERT [121] may be useful in this context, to investigate media bias is a relatively new field of research. Models such as BERT [121], RoBERTa [127], and DistilBERT [167] are pre-trained by Spinde et al. [188], where headlines from news items covering a wide range of political ideologies were subjected to Direct Supervision Learning, and then refining them on BABE [188]. The top-performing models they developed for BERT and RoBERTa, with F1 values of 0.804 and 0.799, respectively, can distinguish between biased and non-biased texts retrieved from BABE. Spinde et al. [187] use a **Multitask Learning (MTL)** technique to train DistilBERT [167] on multiple datasets related to bias. On a subset of BABE, their best MTL model achieves an F1 score of 0.776. In contrast to the MTL model, a **Wiki Neutrality Corpus (WNC)**-based baseline model, which is trained using a portion of the dataset, achieves better results (F1 = 0.782). According to Spinde et al. [187], improvements might be attributed to the substantial bias correlation in the WNC dataset, giving the model bias-specific information. The pre-training method proposed by Spinde et al. [188] falls short in using datasets that are associated with bias to their fullest extent. Their subsequent work [187] employs a sophisticated MTL architecture that mitigates the impact of WNC's pre-training on the bias classification task.

2.4 Bias Label Reliability and IAA

A persistent challenge in media bias research lies in the subjectivity of bias perception and the resulting variability in human annotation. The reliability of bias-labeled datasets hinges critically on the IAA. One of the most widely accepted statistical measures of IAA is Cohen's Kappa (κ) [39]. In the paper [107], *kappa* values are set in the interval [0.61–0.80], representing substantial agreement, while beyond 0.80 values indicate near-perfect consensus. In the foundational work of Haak and Schaer [74], the Q-Bias dataset has been introduced, which includes articles from the ALLSIDES platform. Their study reports $\kappa = 0.84$ for expert agreement on bias labels (left, center, right), suggesting strong reliability of ALLSIDES labels. Similarly, the NBIAS framework [161] reported κ values in the [0.62–0.68] interval, reflecting consistent annotator agreement in classifying news articles along the political spectrum. On the other hand, Spinde et al. [186, 188] discussed the limitations of datasets used in polarity flipping tasks, emphasizing issues such as small sample sizes, lack of annotator metadata, and ambiguous annotation guidelines.

To overcome such limitations, our work relies on the ALLISIDES dataset, which comprises 12,265 news articles sourced from 20 ideologically diverse news outlets, with explicitly defined polarity labels (left, center, right). Metadata such as themes, article images, and corresponding neutral interpretations are considered additionally. Though ALLSIDES does not publish its κ values officially, its methodology has been independently verified in studies such as Q-bias and NBIAS. Selection of our dataset reflects a conscious decision to balance label credibility, domain coverage, and source diversity, which are the key factors in mitigating data-related biases during the pre-training of LLMs for ideological neutrality tasks. Furthermore, to handle residual noise, our approach leverages contrastive learning and reinforcement learning-based reward optimization to improve robustness in settings with imperfect supervision.

2.5 Domain-Adaptive Pre-Training Approaches

The domain-adaptive pre-training language model [23, 108] is imbued with domain-specific knowledge and utilized across various fields (e.g., BioBERT [108], SciBERT [23], ClinicalBERT [85]). According to Mitra et al. [138], there is just one study that used domain-adaptive pre-training to identify media bias. Various domain-adaptive pre-training methods for BERT are investigated by Sun et al. [194] for use in text categorization tasks including topic, question, and sentiment analysis. When the training data corresponds to the domain of the target task, BERT's enhanced performance on several tasks is a result of its pre-training on data from various fields. Gururangan et al. [73] corroborate the findings of Sun et al. [194] by performing RoBERTa's domain-adaptive pre-training across four distinct target domains (computer science publications, biomedical, reviews, and news) and eight subsequent classification tasks. The F1-score of the model is improved by pre-training RoBERTa on massive volumes of news content from F1 = 0.882 to F1 = 0.886 on a hyperpartisan categorization dataset [98]. The model's performance significantly declines to F1 = 0.764 when used in a domain that is not relevant to the domain of interest (unrelated domain-specific pre-training). Operating on the WNC corpus, our model employs a domain-adaptive pre-training approach and is implemented in a manner similar to that of Sun et al. [194] and Gururangan et al. [73]. Given that the majority of current methods emphasize sentence-level bias detection, we build a sentence-level categorization model while adhering to conventional practices. In contrast to sophisticated yet intricate research on media bias detection [188], we implement a more concentrated and straightforward training approach using a considerable volume of highly bias-related data, anticipating significant enhancements in performance.

2.6 Context-Aware Cross-Modality Attention

Contextual information has been shown to enhance dependency modeling because neural representations capture complex aspects of word use, such as grammar and semantics [37], particularly in attention models [130]. For example, internal contextual representations were added to the query and memory components of a classical attention model by Tu et al. [206] and Zhang et al. [233], respectively. According to their study, global contextual information is crucial for simulating the interdependencies across representations. According to Peters et al. [140], multilayer neural networks perform better when syntactic and semantic circumstances are thoroughly modeled at several levels. Our research emphasizes enhancing self-attention networks via the incorporation of contextual information.

Across a range of tasks, including machine translation [210] and natural language inference [177], the self-attention model has shown its effectiveness in modeling discrete sequences; however, it also has limitations [35]. Yang et al. [226] observed that confining the attention model to a localized region may enhance performance, indicating that the traditional self-attention network model inadequately captures the context of a sequence. Tang et al. [195] discovered that traditional self-attention networks do not fully capitalize on the direct connections among the components. Furthermore, Shaw et al. [175] successfully integrated relative positions into self-attention network models, demonstrating that the traditional model necessitates essential information for modeling relationships between states. External contextual representations that contain preceding source sentences were added to the attention network by Bawden et al. [22] and Voita et al. [214]. Our methods place changes into self-attention networks, removing the need for outside resources while maintaining simplicity and adaptability, in contrast to their approach, which requires the embedding of previous words.

2.7 Ideology Prediction

Classical models (e.g., Naive Bayes, **Support Vector Machine (SVM)**; [168]) and deep learning models (e.g., RNN; [90]) have been employed to forecast ideology across various datasets with available ideology labels [50]. The authors Liu et al. [124] pre-trained a language generator that is based on Transformers to lessen ideological bias in the final output. Given that generative models are less successful than **Masked Language Models (MLMs)** for text classification, our objective is to train MLMs to identify ideological content across diverse domains and tasks. This study develops ideology-driven pre-training goals to include domain knowledge for the identification of ideologies.

Limited research has examined PLMs for comprehending political ideology/philosophy expressed in texts. We have at least three novel aspects to our study. Articles pertaining to the same topic are the focus of the loss used by Baly et al. [18], while the primary objective of our triplet loss is to identify instances of ideological similarity or dissimilarity across publications that make reference to the same story. The research conducted by Baly et al. [18] also shows that the results cannot be applied to languages used by other media or in activities linked to ideology, while our recently added narrative aim may effectively prevent the model from relying on media-specific terminology (e.g., "for the New York Times"). Ultimately, we use BIGNEWS, which comprises over 3 million articles, making it more appropriate for pre-training big models compared to the limited dataset of 35,000 articles employed in [18]. We believe we are the pioneers in the systematic study and dissemination of PLMs for ideology-related research.

2.8 Word Embedding Techniques for Disinformation Detection

Al-Tarawneh et al. [5] and Hauschild and Eskridge [78] showed that CNN and LSTM classifiers, when equipped with pre-trained word embeddings, outperformed traditional Bag-of-Words or TF-IDF-based models. Ilie et al. [87] provided a comparative benchmark using GloVe and FastText embeddings, revealing that the combination of word embeddings and context-aware structures enhances predictive robustness. The work [15] focuses on the Arabic social media context, illustrating the complementary strengths of Word2Vec and FastText in hybrid models. Verma et al. [212] introduced WELFake, a model that fuses Word2Vec and GloVe embeddings with syntactic features. Priyanga et al. [153] extended this notion further by evaluating both word and sentence-level embeddings in a clustering framework using cosine similarity and centrality-based network modeling for semantic discrimination. Truică and Apostol [198] proposed a new approach that uses document embeddings to build multiple models that accurately label news articles as reliable or fake. Liu et al. [120] validated this through NewsEmbed, which leverages TF-IDF weighted Word2Vec and Universal Document Encoder representations to model entire news articles effectively. Samadi et al. [166] introduced a dual-channel architecture that combines GloVe with contextual embeddings from ELMo. Guacho et al. [68] presented a tensor decomposition model for misinformation detection using PARAFAC-generated embeddings. Oriola [141] contributed a comparative evaluation of Word2Vec, LDA-based topic vectors, and n-gram features for fake news detection. Our proposed work has been built upon these foundational studies by extending embedding representations across modalities (text and image) and incorporating them into a multimodal ideology-aware debiasing architecture.

2.9 Transformer-Based Methods for Disinformation Detection

Blackledge and Atapour-Abarghouei [27] proposed a robust two-stage transformer-based framework demonstrating improvement in generalizability of the classifier across multiple datasets. Mewada et. al. [137] highlighted that pre-trained transformer models maintain performance even

with varying styles, domains, and class distributions. Similarly, Gondwe [62] explored transformer-based classifiers for real-time misinformation detection. A hybrid model [160] considers news content and its associated social context using transformer encoder, achieving improved accuracy in identifying fabricated information. These findings are echoed in the study of Malla and Alphonse [133], which investigated the fusion of content and user-driven signals within a BERT-based architecture.

Research has also been progressed toward transformer compression and efficiency. DistilBERT and ALBERT, offer similar performance to BERT [197] while significantly reducing computation time, suitable for low-resource settings or edge devices. Papadopoulos et al. [144] examined synthetic misinformation generation using transformers. Complementarily, studies such as Gundapu and Mamidi [71] and Reshi and Ali [164] evaluated model robustness across varying data modalities. The study of Low et al. [129] introduced an empirical evaluation of transformers' generalization performance across opinionated and factual news datasets. Our proposed framework builds upon these foundations by integrating context-aware fusion, ideological debiasing, and multimodal transformers.

2.10 Sentence Transformer-Based Approaches for Disinformation Detection

Sentence-level transformer models are pre-trained on large-scale corpora and fine-tuned for specific downstream tasks, offering substantial improvements over traditional feature-based or shallow neural network approaches. Al-Quayed et al. [4] introduced a hybrid framework that integrates BERT-based sentence embeddings with linguistic features. Vijjali et al. [213] proposed a two-stage pipeline for COVID-19 misinformation detection. Truică et al. [202] incorporated BiLSTM layers on top of sentence transformer embeddings (BART and XLM-R) to address multilingual fake news detection. Kasnesis et al. [97] conducted a comparative analysis of transformer models for stance and subjectivity classification.

Vaibhav et al. [209] utilized BERT embeddings within a sentence-level graph structure to capture semantic relationships across documents. Complementary to this Tufchi et al. [207] introduced a fusion model combining sentence transformer embeddings with Variational Autoencoders and topic modeling to learn hierarchical representations of disinformation content. Similarly, the exBAKE [96] demonstrated the superiority of domain-adapted BERT-based representations over conventional methods by addressing data imbalance and optimizing loss functions for better generalization. Gundapu and Mamidi [71] employed an ensemble of BERT, ALBERT, and XLNet for COVID-19 fake news detection. Another study by Raza and Ding [160] leveraged a joint representation of news articles and social discourse, which integrates user comments and meta to improve disinformation classification performance. In our proposed framework, we extend this line of research by incorporating contextual sentence embeddings into a cross-modal setup, bridging textual and visual signals, and aligning them with ideology-aware debaising mechanisms.

2.11 Complementary Modeling Strategies in Disinformation Detection

2.11.1 Mixture-of-Experts (MoE) Approaches. MoE models aim to route inputs through specialized "expert" subnetworks, thereby improving generalization and reducing overfitting. Petrescu et al. [148] proposed a true Mixture-of-Transformers framework where multiple transformer experts were assigned based on the input language. Similarly, Piergiovanni et al. [150] presented a task-level MoE setup, where a unified transformer was pre-trained on a diverse set of cross-modal and object-aware tasks, such as captioning.

Hayase et al. [79] offered a data-centric perspective, showing that pre-training on mixed corpora (e.g., Reddit, Wikipedia) leads to emergent expert behaviors even in non-modular models. Bansal et al. [19] developed an Indic-language abuse detection system using a shared multilingual

transformer backbone fused with BiGRU and emoji expert embeddings. Together, these works emphasize that MoE whether architecturally modular, task-based, or data-driven, offers critical advantages for misinformation and disinformation detection in complex, multilingual, and multimodal contexts.

2.11.2 Ensemble-Based Detection Architectures. Ensemble learning techniques have proven to be powerful for disinformation detection by combining complementary views of content, context, and propagation. Gul et al. [70] proposed an ensemble of BiGRU and **Graph Neural Network (GNN)** components to jointly model sequential language patterns and relational dependencies, achieving high accuracy on LIAR and FakeNewsNet datasets. Similarly, Malik et al. [132] developed an ensemble GNN framework that fuses user engagement metadata with content embeddings (TF-IDF, BERT), using soft voting across multiple graph backbones (GCN, GAT, R-GCN).

Truică et al. [200] introduced GETAE, a deep neural ensemble that combines BiLSTM-based semantic features with GCN-based structural embeddings using an attention-gated late fusion mechanism. Building on this, Koloski et al. [101] proposed a heterogeneous representation ensembling model that fuses knowledge graphs, contextual embeddings, and propagation signals through meta-level classifiers. Xie et al. [222] further expanded this paradigm by integrating HetGNNs with knowledge graphs and user-post propagation trees. Dar et al. [42] introduced a knowledge graph fusion model that unified content-based and evidence-based features through semantic aggregation. Their ensemble approach achieved state-of-the-art results during the COVID-19 infodemic, particularly on multilingual and noisy data.

2.11.3 Social Network and Propagation-Based Methods. An emerging body of research has explored the detection of fake news using features derived from user behavior, credibility cues, and propagation patterns. Roy et al. [165] proposed a hybrid deep learning model incorporating speaker metadata, such as political affiliation and historical credibility, to enhance classification over the LIAR dataset. Truică et al. [199] further extended this idea by introducing DANES, a dual-channel deep ensemble framework that fuses textual embeddings with user engagement signals using self-attention to dynamically weigh social influence. Ali et al. [7] focused on temporal social behavior by employing LSTM, BiLSTM, and GRU models to process sequential post activity. Venkatachalam et al. [211] introduced DeepFND, a CNN-LSTM ensemble enriched with user behavior data such as likes, shares, and comment counts. Almandouh et al. [49] proposed an ensemble that integrates user profile metadata, such as verification status, retweet frequency, and audience size. Aslam et al. [14] designed a two-stream architecture to separately process content and speaker metadata to infer latent trustworthiness and intent. Agarwal and Dixit [2] extended this by introducing author credibility scoring, aligning subject themes with author profiles to enhance trust modeling without explicitly relying on network graphs.

2.12 Polarity Debiasing

Researchers in the field of political science have examined how changing the source attributions of the articles might affect how people see the news. The same news content was experimentally distributed to a number of media outlets with different perceived political philosophies, including FOX and CNN [89]. These studies show that perceptions of bias in the content may be greatly influenced by heuristics, such as the media's source. Although the majority of these studies have focused on how media sources affect perceived bias, some social scientists suggest that material presentation be changed further to find additional subtle markers of perceived bias [21]. Our research offers an additional approach to control the presentation of material using computer-assisted techniques.

There are a lot of technical similarities between our methods for inverting media bias and the general-purpose style transfer strategies. The *style* in our technique denotes the polarity label

(*left-oriented*, *center-oriented*, or *right-oriented*), and attaining the transfer of style while preserving substance is our objective. Shen et al. [176] increase the system's capacity to discriminate between style and content distributions by including adversarial networks into a cross-aligned auto-encoder architecture. In order to ensure that the style and content spaces are clearly separated during training an encoder–decoder network, John et al. [94] used adversarial loss and auxiliary multitask loss together. They do this by meticulously dividing the latent representation into style- and content-related domains. The concept of style-content separation has been the basis for several suggested techniques [54, 106]. Style transfer techniques use either reinforcement learning or sequential transfer to enhance the generating process. Wu et al. [219] have forth a hierarchical reinforcement sequence operation method for passing on styles. Similarly, Wu et al. [221] use a simple but effective technique that does not involve creating a new sentence from scratch, but only replaces the target sentimental expressions with the original emotional markers in the phrase. Beam search and multinomial sampling are integrated throughout the rewards estimate process by Gong et al. [64]. They first generate a reference target sentence using beam search, and then they execute the subsequence using multinomial sampling at each iteration to sample whole phrases. Liu et al. [125] change the political polarity of news articles in two steps. Initially, they develop a word embedding model that is cognizant of subject and ideological attributes, then use a refined simulated annealing technique to choose appropriate words for polarity reversal. This method's practical applications are limited since it relies on the premise that surface-level, or token-level, recognition of polar text is possible, but it cannot negate the media bias based on ideological polarity.

Our methodology parallels that of Chen et al. [35], which seeks to invert the bias seen in news headlines. They use a cross-aligned auto-encoder, which is comparable to the method used by Shen et al. [176], trained on articles of opposing ideologies to produce inverted titles. The authors acknowledge that one disadvantage of their research is the potential loss of content information. The produced headlines and the ground-truth headlines have very little content overlap, indicating that the auto-encoder approach tends to exclude excessive non-polar information. However, our approaches are not predicated on the notion that latent space may be used to properly discern between style and content. In the event that any content-related data loss occurs due to the artificial separation of content and style, we include an attention mechanism into the transformer block and style embedding, especially polarity embedding, to implicitly encode both content and style information [210]. Additionally, we expand the goal to include depolarizing the whole article instead of just the headlines, which poses a more difficult task since it requires maintaining grammatical and contextual integrity across the entire text.

3 Pre-Training Datasets

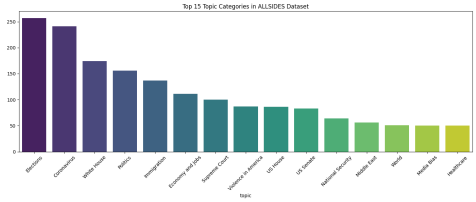
The idea and classifications of media framing bias are described in the literature on NLP and social science research [51, 57]. The selective presentation of facts intended to influence readers' attitudes is referred to as *Informational framing bias*. The phrase *Lexical framing bias* refers to a way of writing or language that might mislead readers. Nevertheless, the term alone is insufficient to comprehend the precise manifestation of framing bias in concrete instances, such as the ALLSIDES dataset.

3.1 ALLSIDES

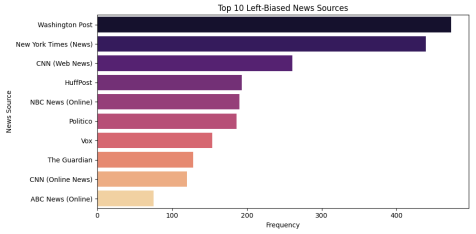
ALLSIDES [179] offers triplets of news items, including coverage of the same event by right-wing, centrist, and left-wing American periodicals, accompanied with a title and unbiased summary provided by an expert. The neutral overview enables readers to get an unbiased comprehension of a subject prior to engaging with specific articles. The three most common topics are "White House," "Elections," and "Politics." For our task, the neutral article summary is the objective output A_{neu},

Table 2. Overview of the ALLSIDES Dataset and the Class Distribution

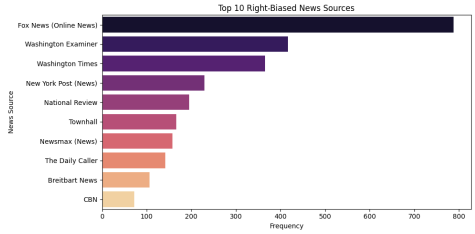
Class Labels	No. of Samples
Left-Oriented News	3,067
Right-Oriented News	3,067
Center-Oriented News	3,067
Neutral/Roundup News	3,067



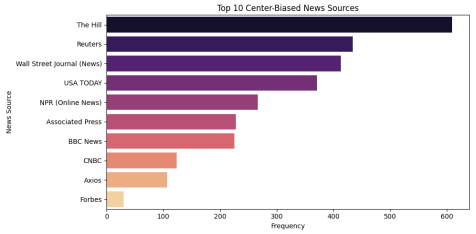
(a) Top 15 Topic Categories in ALLSIDES Dataset



(b) Top 10 Left-Biased News Sources



(c) Top 10 Right-Biased News Sources



(d) Top 10 Center-Biased News Sources

Fig. 2. EDA on ALLSIDES dataset focusing on topic distribution and media outlet frequencies.

and the article triplets are the source inputs $\{A_L, A_C, A_R\}$. The term "center" does not imply absolute impartiality, as shown in *Supplementary Materials; Section A, Table 1*; instead, "center" media outlets are comparatively less affiliated with a certain political viewpoint. Furthermore, the title triplets $\{T_L, T_C, T_R\}$ and the neutral issue title T_{neu} are also collected along with the meta-information, like headlines, date, photos, subject tags, and media name. There are total 3,067 triplets (12,268 articles), as shown in Table 2, of which $\frac{2}{3}$ (2,045) are used for the training and validation set in a 80 : 20 ratio, while the remaining (1,022) are the test set. The stratified K-fold cross-validation method ($K = 5$) was also employed to maintain the statistical consistency of category representation.

3.2 ALLSIDES: Exploratory Data Analysis (EDA)

To better understand the structure and biases inherent in the ALLSIDES dataset, we performed an in-depth EDA, focusing on topic distribution and media outlet frequencies, shown in Figures 2 and 3. To further characterize the structure and ideological nuances embedded in the ALLSIDES dataset, we conducted additional analyses focusing on sentiment polarity, subjectivity, and inter-bias content similarity across the triplet structure (left, center, and right perspectives), given in Figure 4. We examine the stylistic and thematic variations across ideologically diverse news narratives in the ALLSIDES dataset and conduct additional analyses focusing on linguistic prominent (via TF-IDF), and semantic themes (via word clouds), shown in Figures 5 and 6.

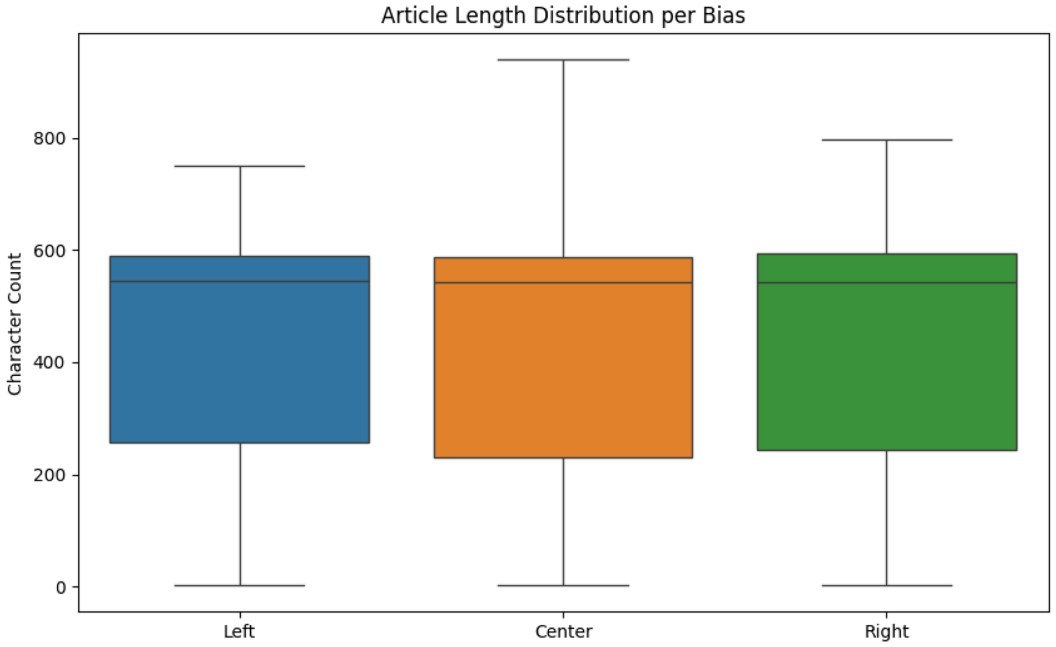


Fig. 3. Article length distribution per bias.

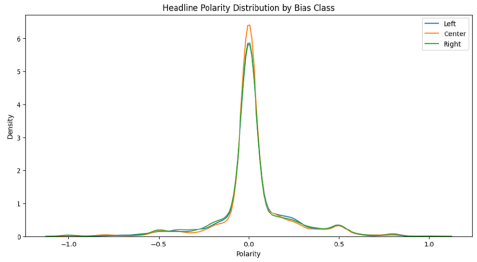
3.2.1 Description of Data Distribution. We found that the dataset predominantly covers politically sensitive themes (Figure 2(a)). The top recurring topics include *Elections*, *White House*, *Supreme Court*, *Immigration*, and *Gun Control*. These categories reflect the typical domains where ideological bias is most pronounced, supporting the relevance of the dataset for studying political slant in media narratives.

A breakdown of news sources across the ideological spectrum confirms the inclusion of diverse media organizations. Left-leaning articles commonly originate from sources such as *Vox*, *CNN*, and *NBC News* (Figure 2(b)). Center-oriented document is predominantly drawn from *BBC News*, *Associated Press*, and *Axios* (Figure 2(d)), while right-leaning articles are frequently attributed to *Fox News*, *The Blaze*, and *Breitbart* (Figure 2(c)). This distribution supports the dataset's design of capturing viewpoint plurality.

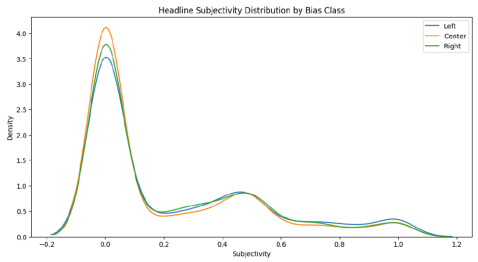
We examined the length of article bodies across left-, center-, and right-biased classes, as shown in Figure 3. On average right-biased articles tend to be *slightly longer*, while center and left articles have more *compact distributions*. This stylistic variation may carry implications for both content complexity and framing strategies, which future bias models can exploit.

3.2.2 Headline Sentiment Analysis. Using TextBlob, we analyzed the sentiment polarity and subjectivity of the headlines across bias classes, given in Figures 4(a) and 13(c), respectively. We observed that left-leaning headlines, on average, displayed *slightly higher polarity scores*, suggesting a marginally more *positive tone*. In contrast, right-leaning headlines were found to be more *neutral in polarity* and notably *lower in subjectivity*, indicating a *declarative or factual tone*. Center-oriented headlines occupied an *intermediate position*. These stylistic differences reflect how framing bias may manifest through emotional cues and assertiveness in headlines.

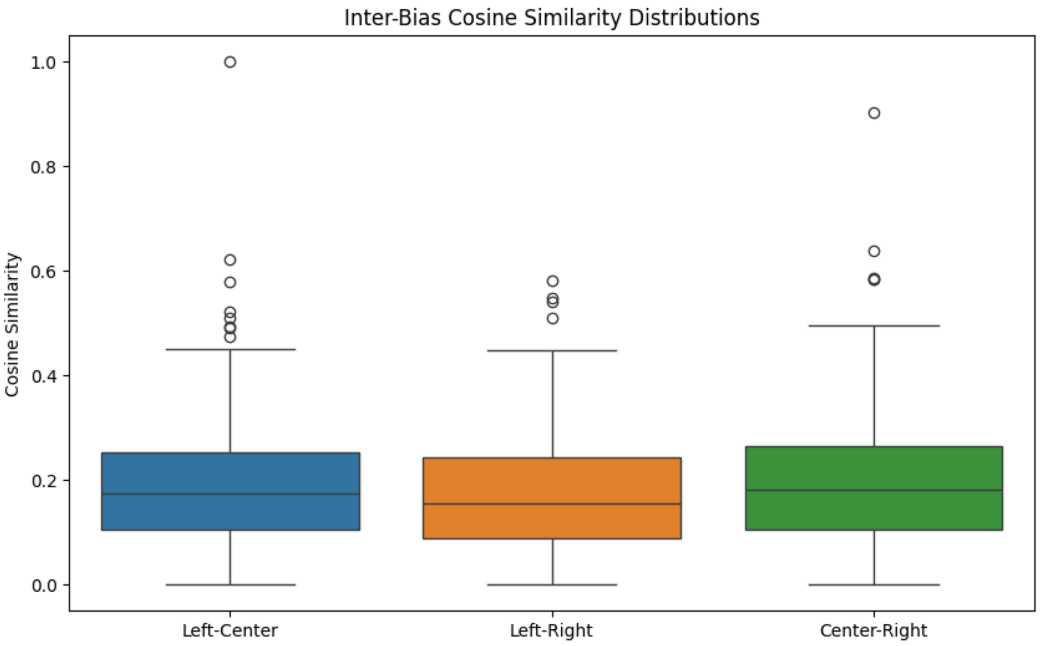
3.2.3 Inter-Bias Article Similarity. To quantify the degree of overlap among articles presenting different ideological views of the event event, we computed pairwise cosine similarity scores



(a) Headline Polarity Distribution by Bias Class



(b) Headline Subjectivity Distribution by Bias Class

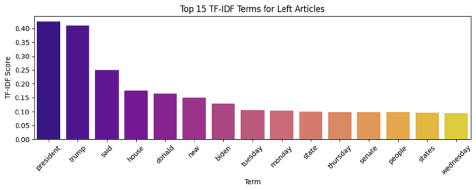


(c) Inter-Bias Cosine Similarity Distributions

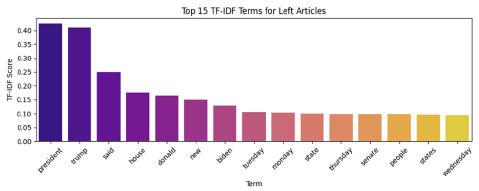
Fig. 4. EDA on ALLSIDES dataset focusing on semantic polarity, subjectivity, and inter-bias content similarity across the triplet structure (left, center, and right perspectives).

between left, center, and right articles using TF-IDF representations, as shown in Figure 4(c). Based on a stratified sample of 500 triplets, we report the following mean similarity scores of *Left-Center*: 0.186, *Center-Right*: 0.191, and *Left-Right*: 0.172. These scores confirm that left and right leaning articles are the most *dissimilar*, while center articles maintain a relatively *balanced linguistic* and *topical overlap* with both extremes. This validates the dataset's triadic design and supports the motivation behind using center-oriented content as a target in our bias mitigation framework.

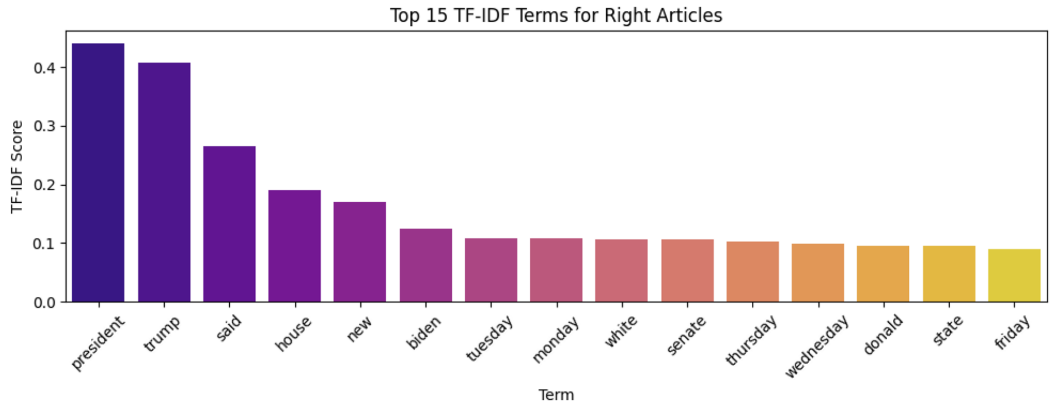
We extracted the top TF-IDF weighted terms for left, center, and right leaning articles using a corpus level analysis of unigrams, as shown in Figure 5(a)–(c), respectively. As anticipated, left leaning articles frequently featured terms such as *equity*, *climate*, and *justice*, reflecting a focus on social issues and environmental advocacy. Right-leaning articles emphasized terms like *border*, *enhancement*, and *tax*, consistent with economic and security-oriented discourse. Center articles surfaced a blend of analytical and administrative terms including *policy*, *economy*, and *moderate*.



(a) Top 15 TF-IDF Terms for Left Articles



(b) Top 15 Terms for Center Articles



(c) Top 15 Terms for Right Articles

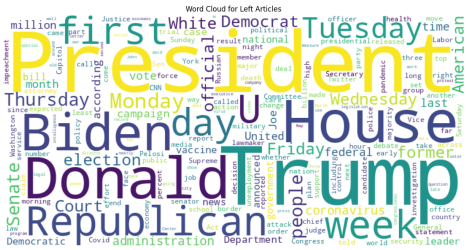
Fig. 5. EDA on ALLSIDES dataset focusing on linguistic prominence (via TF-IDF).

These differences align well with known ideological leanings and offer valuable signals for automatic bias identification and content framing studies.

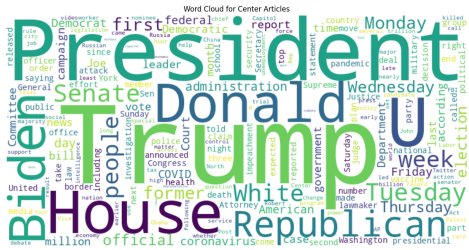
To visually capture the dominant lexical choices within each ideological cluster, we generated word clouds from the article bodies (Figure 6(a)–(c)). The results revealed that left-oriented texts emphasized *progressive sociopolitical themes*, right-oriented articles concentrated on *nationalistic* and *regulatory language*, while center-oriented articles exhibited *balanced narrative* structures with *policy-level terminology*. These visualizations underscore how lexical choices can serve as both explicit and implicit carriers of ideological stance.

3.3 WNC

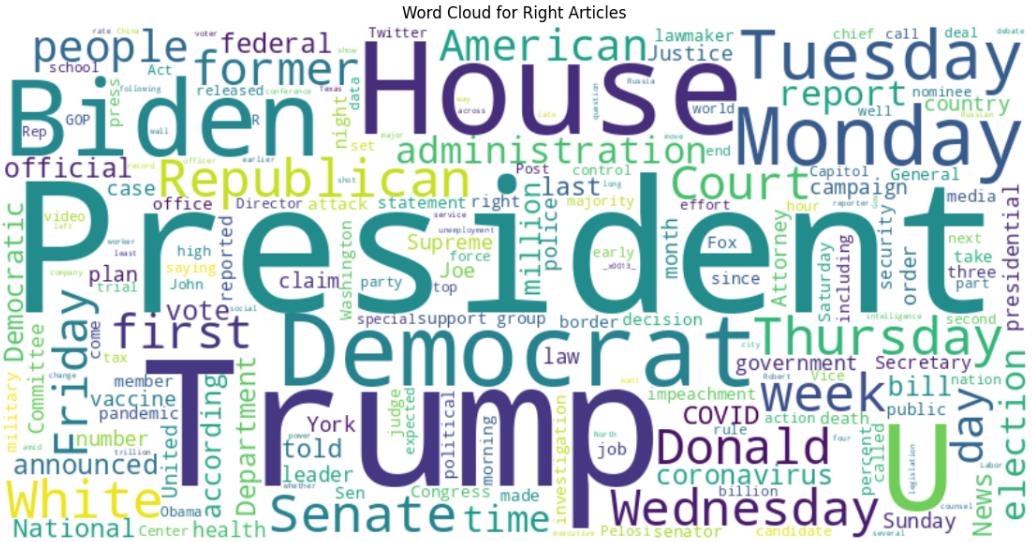
Aligned sentences that have been *pre-* and *post*-neutralized by English Wikipedia editors make up the WNC [154] (*Supplementary Materials; Section B, Table 2*). Regular expressions are used to analyze 423,823 Wikipedia modifications from 2004 to 2009, whereby editors provide NPOV-related justifications [163]. Using a sliding window of size k = 5, the *pre-* and *post*-text sentences are synchronized by calculating pairwise BLEU scores [145] between sentences. The sentences with the highest scores are then aligned. The data shown in Table 3 pertain to the corpus statistics. The final data includes (1) a parallel corpus with 180,000 in prejudiced remarks and their neutral equivalents, and (2) along with each altered statement, there were 385,000 neutral sentences that the editor did not alter.



(a) Word Cloud for Left Articles



(b) Word Cloud for Center Articles



(c) Word Cloud for Right Articles

Fig. 6. EDA on ALLSIDES dataset focusing on semantic themes (via word clouds).

Table 3. WNC Corpus Statistics

Data	Sentence Pairs	Total Words	Seq Length (Mean)
Biased-Full	181,496	10.2M	28.21
Biased-Word	55,503	2.8M	26.22
Neutral	385,639	17.4M	22.58

3.4 BIGNEWS

The BIGNEWS [128] dataset is compiled from internet news items exhibiting a variety of ideological perspectives and linguistic styles. Eleven media sources have been classified into three ideological categories: left, center, and right. All content generated by these sources from January 2000 to June 2021 has been extracted from Common Crawl and the Internet Archive. The dataset includes a total of 3,689,229 news articles related to US politics. To alleviate the bias resulting from certain ideologies dominating model training, the corpus is down-sampled to ensure equal contribution from each ideology. The down-sampled corpus, BIGNEWSBLN, consists of 2,331,552 news stories, as

Table 4. Data Regarding BIGNEWSBLN

	Daily Kos	Huffington Post	CNN	The Washington Post	The New York Times	USA Today	AP	The Hill	The Washington Times	FOX	Breitbart
Ideology	L	L	L	L	L	C	C	C	R	R	R
# Articles	100,828	241,417	64,998	198,529	173,737	170,737	279,312	322,145	243,181	330,166	206,512
# Words	738.7	729.9	655.7	803.2	599.4	691.7	572.3	426.3	522.7	773.5	483.5

Media outlets are categorized by ideology into three distinct groups: left (L), center (C), and right (R), according to the ALLSIDES and media bias chart.

Table 5. When Evaluating PLMs, Several Datasets Are Used, Each with Its Own Unique Text Type, Length, Training Set Size (# Train), and Criteria for Partitioning

Data	Genre	# Train	Length	Split
Congress Speech [58]	Speech	7,000	538	Random
Hyperpartisan [98]	News	425	556	Random
ALLSIDES [179]	News	7,878	863	Time
YouTube User [220]	Cmt	1,114	1,213	User
YouTuber Cmt [220]	Cmt	6,832	197	User
BASIL-Article [51]	News	450	693	Story
BASIL-Sentence [51]	News	1,197	27	Story

Time split alludes to the process of training on historical data and testing on subsequent data. The story split categorizes items based on narrative clusters. The user split signifies that the individuals in the test group were not present in the training data.

detailed in Table 4. The dataset is then down-sampled, and the articles are categorized into clusters, each including items aligned with the same narrative. The dataset produced, BIGNEWSALIGN, comprises 1,060,512 entries featuring aligned articles.

3.5 Dataset Transparency

One crucial concern in large-scale ideological bias detection tasks is the quality and consistency of human annotations, especially given the subjective nature of political stance interpretation. We incorporate the concept of Cohen's Kappa (κ) [39] as a standard statistical metric for evaluating IAA. Cohen's Kappa quantifies the level of agreement between two annotators while correcting for chance agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{1}$$

where p_o represents the observed agreement between annotators, and p_e denotes the expected agreement by chance. The κ value ranges from -1 (complete disagreement) to 1 (perfect agreement), with 0 indicating chance-level agreement.

Based on the results of [107, 161], we underscore strong consensus in labeling partisan content and the usability of the ALLSIDES dataset for training robust bias detection models, despite the inherent interpretative nature of ideological annotation. For the BIGNEWS dataset, while Cohen's Kappa scores are not explicitly reported, we have followed their labeling protocol and exercised caution applying our domain-aware supervision, contrastive triplet-based learning strategies and regularization-based training objectives to reduce the model's sensitivity to potential annotation noise.

3.6 Other Corpora

Congress Speech [58]. The Congress Speech comprises remarks from US legislative records, categorized as either liberal or conservative.

Media Cloud. Harvard and MIT are at the forefront of an academic research effort called Media Cloud. Articles from many media sources are compiled by the Media Cloud. Approximately 360,000 full-length articles were gathered from Media Cloud throughout the period from May 1, 2018, to May 1, 2019. The pieces originate from 22 media sources. Based on data from the Pew Research Center, each source is assigned a polarity label that indicates ideology. Pew's data derives from a poll examining news consumption among individuals with varying political affiliations. Five types of people are identified by the Pew study: *consistent liberals*, *mostly liberals*, *mixed*, *predominantly conservatives*, and *consistent conservatives*. Pew assigns the labels based on responses to questions regarding different political views. Polarity labels are *liberal* (i.e., Pew's *consistent liberal* and *mostly liberal*), *neutral* (i.e., Pew's *mixed*), and *conservative* (i.e., Pew's *consistent conservative* and *mostly conservative*). This dataset is utilized for the polarity inversion of news articles.

Hyperpartisan [98]. Hyperpartisan refers to the classification of an article as either hyperpartisan or not. The dataset has been transformed into a three-class classification job by categorizing hyperpartisan news into left and right.

YouTube [220]. The YouTube dataset comprises topics pertaining to YouTube. The terms *cmt.* and *user* indicate left/right predictions at the user-level and comment-end, respectively.

Twitter [152]. It is a collection of Twitter users who were grouped into three different groups based on their self-reported ideologies, which were then ranked on a 7-point scale.

Bias Annotation Spans on the Information Level (BASIL) [51]. *BASIL* is a dataset regarding media bias that was released by Fan et al. [51]. Three articles per event, totaling 300 news items related to 100 occurrences, make up the dataset. The three articles, which represented right-, neutral-, and left-leaning platforms, respectively, were taken from Fox News, the New York Times, and the Huffington Post.

3.7 Evaluation Metrics for Imbalanced Datasets

Imbalanced data distributions pose a critical challenge in supervised learning, particularly in the context of ideological bias detection. Traditional metrics such as overall accuracy fail to provide an adequate assessment in such scenarios, as they are skewed by majority-class dominance and may obscure poor performance on underrepresented ideological categories [31]. To mitigate this issue, we adopt the macro-averaged Precision ($\mathcal{P}_{\text{macro}}$), Recall ($\mathcal{R}_{\text{macro}}$), and F1-Score ($\mathcal{F}_{\text{1-macro}}$), computed as follows:

$$\mathcal{P}_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{TP_k + FP_k}, \quad \mathcal{R}_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{TP_k + FN_k}, \tag{2}$$

$$\mathcal{F}_{\text{1-macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{2\cdot\mathcal{P}_k\cdot\mathcal{R}_k}{\mathcal{P}_k + \mathcal{R}_k}, \tag{3}$$

where k is the number of classes, and TP_k, FP_k, and FN_k denote the true positives, false positives, and false negatives for class k, respectively. These metrics ensure that each class contributes equally to the final evaluation, regardless of its prevalence in the dataset. In addition, we report *Balanced Accuracy ($\mathcal{BA}$)*, defined as the average of class-wise true-positive rates (sensitivity):

$$\mathcal{BA} = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{TP_k + FN_k}, \tag{4}$$

$\mathcal{BA}$ is equivalent to macro-recall and provides a clearer view of model robustness in minority-class detection. Li et al. [112] and Krawczyk et al. [103], argue that AUC-PR is a superior alternative to AUC-ROC in heavily skewed distributions, as it avoids being dominated by true negatives. In the article, we include the **Area Under the Precision-Recall Curve (AUC-PR)**. Our experimental framework integrates *macro-F1*, *balanced accuracy*, *class-wise recall*, *AUC-PR*, and *Kappa score* to deliver a comprehensive and imbalance-aware evaluation protocol. These metrics are motivated by best practices from decision tree ensembles [31], hesitant fuzzy systems [169], and cost-sensitive frameworks [103, 112], ensuring both statistical rigor and practical relevance in real-world media bias detection scenarios.

Additional strategies have been proposed to further mitigate imbalance-induced bias. For example, decision tree-based methods have been adapted to handle class imbalance via robust split criteria and tree pruning heuristics, as demonstrated by Liu et al. [126] and Truică et al. [205]. These approaches modify entropy and Gini-based splitting thresholds to favor minority class separability. Moreover, algorithmic innovations in multiclass settings have been explored by Ryan Hoens et al. [81], who introduce structure-preserving multiway decision trees designed to combat skew in multiclass fake news contexts. Finally, ensemble-based learning frameworks have been shown to improve classification on imbalanced data, particularly when integrated with fake news detection modules, as shown in Agarwal and Dixit [2].

4 Proposed Approach: CANDOR

The primary goal of our proposed model is to smoothly integrate multimodal knowledge into the LLM architecture. Here we introduce the model CANDOR, which consists of a Context-Aware Cross-Modality Attention and an Information Aggregator. Given the textual biased news as input along with the associated image, the former, i.e., the Context-Aware Cross-Modality Attention, aptly introduces multimodal information and political ideology/philosophy in the textual representations, while the latter conglomerates the image-ideology-information infused textual representations.

4.1 Polarity Detector

A polarity detector is essential for identifying phrases with polarity, particularly when the source material consists of full-length articles rather than concise headlines. A detector must be able to identify polar text in various contexts and accommodate input of any length. Current methodologies either neglect contextual information by choosing words within static contexts [56] or exhibit constrained modeling capability with lengthy sequence inputs [90] (RNN-based approaches). Our methodology for the automated identification of polarity in text utilizes the efficacy of domain-adaptive pre-training via an unsupervised language model trained on an extensive corpus.

The bias domain (WNC) is used for pre-training neural-based language models and ALLSIDES is utilized to evaluate them on the media bias classification challenge, as shown in Figure 7. We create a learning task based on ALLSIDES, which is then optimized (Section 4.1.1). The next step is to choose suitable transformer models and set their starting weights using pre-trained weights (Section 4.1.2). We train the models on the WNC to adapt them for the media bias domain (Section 4.1.3). Ultimately, all models have been refined and assessed on ALLSIDES. We anticipate that by comparing word representations to data distributions from biased and neutral news sources, domain-adaptive pre-training will enhance word representations.

4.1.1 Learning Task. Intermediate training helps to improve the language models. For every $i = 1, ..., N$ in our corpus X, we have N phrases $x_i \in X$ that have ternary bias labels encoded as 1 and 0, representing left-oriented, center-oriented, and right-oriented bias. The goal is to assign the

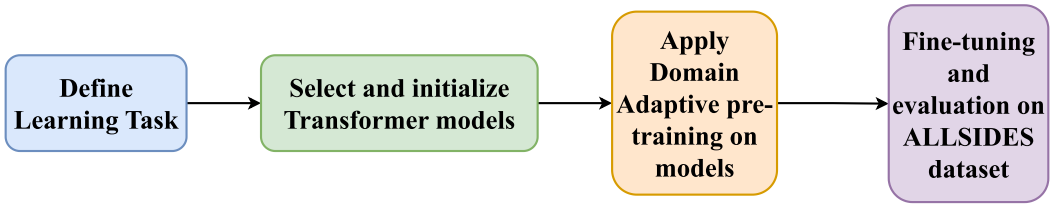


Fig. 7. Pre-training and fine-tuning workflow.

correct label $y_i = \{0, 1\}$ to x_i. Slashing cross-entropy loss is the objective of training:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=\{0,1\}} y_{i,k}\log(\hat{y}_{i,k}), \tag{5}$$

where $\hat{y}_{i,k}$ represents the anticipated score for a phrase by the model, whereas $y_{i,k}$ is the genuine ternary label.

4.1.2 Transformer-Based Models. Concerning our domain-specific pre-training, we chose BERT [45] and RoBERTa [127] since they were identified as the best models in Spinde et al. [188]. By doing this, we get the highest level of compatibility with previous top-tier bias classifiers. We also include T5 and BART since encoder–decoder architectures outperform BERT in certain NLP applications, such as GLUE [118]. In our media bias detection challenge, we choose appropriate models to examine the effectiveness of combining autoencoder and autoregressive components (BART) with advanced MTL architectures (T5).

By maximizing an unsupervised learning problem focused on *Next Sentence Prediction* and *Masked Language Modeling*, BERT is able to obtain bidirectional word representations from unlabeled text. With regard to pre-training data, RoBERTa differs somewhat from BERT and does not include the Next Sentence Prediction objective. BART employs text modifications via the introduction of noise and acquires representations by recreating the initial text sequence. T5 accomplishes all of its training objectives via a text-to-text approach and uses an MTL architecture that has been pre-trained on several supervised and unsupervised activities. Every model is customized for the media bias domain and tested on the media bias classification task at the phrase level (Section 6).

4.1.3 Domain-Adaptive Pre-Training. According to Beltagy et al. [23], and Lee et al. [108], it is essential to adjust a language model that has already been trained for a certain domain if the target data is substantially different from the training data. Media bias, characterized by tendentious and doubtful terminology in biased news, differs from the majority of domains on which BERT-like models are pre-trained. Information from online text databases, news databases, and commonsense reasoning are all part of RoBERTa. In addition to the English Wikipedia and the BooksCorpus, which BERT is trained on. A large number of BERT models have already undergone training on political orientation classification and fake news [27]. In spite of this, media bias is quite different from the ideas of fake news and political orientation.

We use the WNC in our domain-adaptive pre-training to make the learning task, specified in Section 4.1.1, better. The optimal method for adapting to a particular domain and task is domain-adaptive pre-training, as stated by Sun et al. [194] and Gururangan et al. [73], which is supported by our technique. Since both WNC (pre-training) and ALLSIDES (fine-tuning) include manually annotated phrases and have comparable bias patterns, we anticipate that our fine-tuning outcomes will be enhanced by implementing the proposed pre-training task.

Initially, we transform the input sequence in accordance with the methodology outlined in the original RoBERTa implementation [127]. This includes two unique tokens, [SEP] and [CLS], toward the end of the sequence. The sequence classification model was then fed the altered data. The [CLS] acts as a bridge, linking the linear layer to the foundational attention layer. The categorization of the [CLS] token can thus be determined by comparing its attention score to that of all other tokens in the sequence. We append a linear layer to the language model to enable sequence classification across three categories: *left-oriented*, *center-oriented*, and *right-oriented*. The maximum sequence length was established at 1,024 tokens, sufficient to include the majority of articles in our collection. The influence of specific tokens x_i on the categorization outcome ($Y = y$) may be represented as:

$$\begin{aligned} P\left(Y = y|x_i\right) &= \sum_{h_i \in heads} Attn(x_i, [CLS])_{h_i}, \\ P\left(Y = y|x_i\right) &= \sum_{h_i \in heads} softmax\left(\frac{Q_{x_i} \cdot K_{x_i}^T}{\sqrt{d_k}}\right) V_{[CLS]}, \end{aligned} \tag{6}$$

where the hidden dimension is denoted by d_k and the query and key vectors are denoted by Q, K. We calculate the overall attention scores across all the heads for [CLS] and other tokens, where $Attn$ is the multihead attention mechanism.

4.2 BIASLENS

We show our ongoing pre-training techniques grounded on the suggested *ideology/philosophy objective*, which enhances representation learning to more effectively differentiate ideological content by analyzing articles on the same topic (Section 4.2.1). A story/narrative objective assists in honing concentration, which in turn enhances the ideological content, introducing BIASLENS, which stands for Bias-aware Ideology and Article Story-driven Learning with Explicit Negative and Same-story contexts (Equation (9)).

4.2.1 Ideology-Driven Pre-Training Objectives. We employ BIGNEWSALIGN, which sorts articles according to stories in order to provide the narrative level context to train the model, in order to improve representation learning that successfully incorporates ideological content. We use triplet loss [172], which functions on *triplets* of $<anchor, positive, negative>$, to promote proximity between anchor and positive representations while distinguishing the anchor from negative samples.

Our ideology/philosophy objective, which is our primary pre-training goal, uses triplet loss as a means of training the model to produce ideologically biased representations by comparing and contrasting sections of the same story written by media with various views. In Figure 8, we choose a publication from either left-leaning or right-leaning media as the *anchor* for a certain story cluster. Articles having the same ideology and belonging to the same cluster are referred to as *positive* samples, whereas those with contrasting ideologies are classed as *negative* samples. The purpose of the ideology is articulated as follows:

$$\mathcal{L}_{ideology} = \sum_{t \in \tau_{ideology}} \left[1 - \cos\left(t_{anchor}, t_{positive}\right) + \cos\left(t_{anchor}, t_{negative}\right) + \delta_{ideology}\right]. \tag{7}$$

In the triplet t, $\tau_{ideology}$ denotes the collection of all ideology triplets, whereas t_{anchor}, $t_{positive}$, and $t_{negative}$ represent the [CLS] embeddings of anchor, positive, and negative articles, respectively. Furthermore, $\delta_{ideology}$ functions as a hyperparameter.

Subsequently, by adding a narrative objective to the ideological target, we enable the model to concentrate on semantically meaningful content. The model offers its approach, termed "shortcuts," which includes media-specific terminologies, to identify ideology. Figure 8 shows that in order to create the story triplet, we examine articles from the same media source that cover various themes

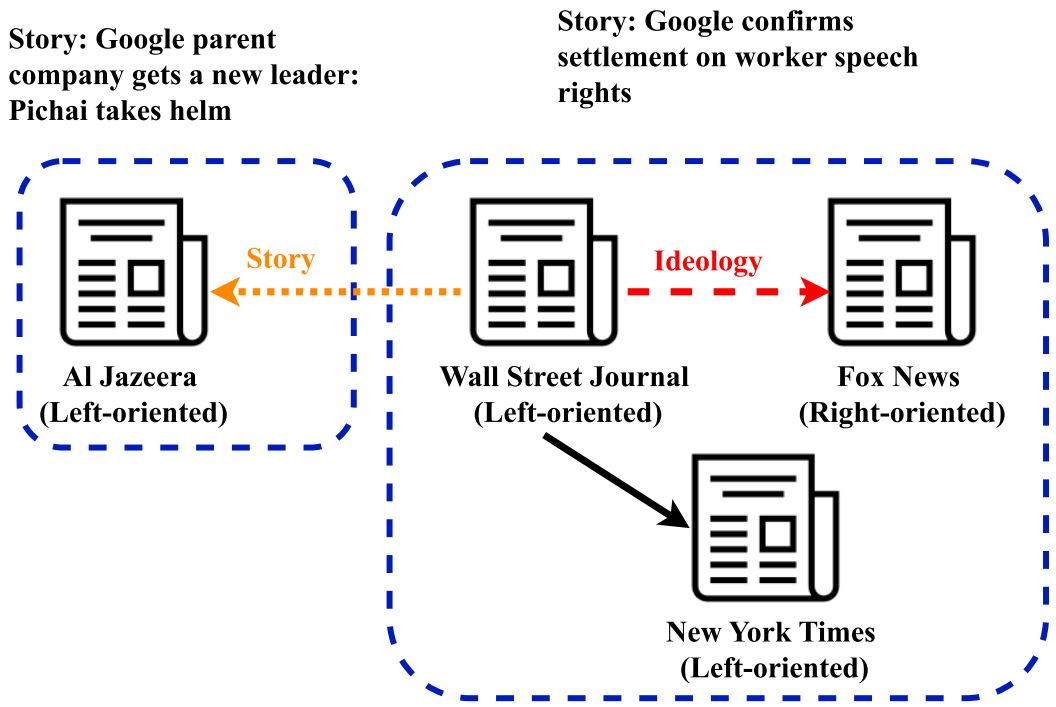


Fig. 8. Development of the ideology and narrative objective. In this case, the anchor is the main Wall Street Journal story itself. For both objectives, a positive pair connection is shown by the solid black arrow; for the ideology objective, the red dashed arrow denotes a negative pair connection; for the story/narrative objective, the orange dashed arrow indicates a negative-pair connection.

as negative examples, but adhering to the same $<anchor, positive>$ pairings as in the ideology triplet. Correspondingly, our narrative aim is articulated as follows:

$$\mathcal{L}_{story} = \sum_{t \in \tau_{story}} \log\left(1 + e^{\|t_{anchor} - t_{positive}\|^2 - \|t_{anchor} - t_{negative}\|^2}\right) + \delta_{story}, \tag{8}$$

where the hyperparameter δ_{story} is adjusted based on the validation set, and τ_{story} includes all story triplets. Here are our final pre-training goals, which are a combination of the aforementioned:

$$\mathcal{L} = \beta \times \mathcal{L}_{ideology} + (1 - \beta) \times \mathcal{L}_{story}, \tag{9}$$

where $\beta = 0.2$. $\mathcal{L}$ is used to generate BIASLENS via ongoing RoBERTa training [127].

4.3 Media Bias Neutralizer (Ideological Polarity Debiasier)

LLMs have shown positive outcomes in a number of natural language generating applications and are capable of producing text that closely resembles human writing [229, 239]. One such application is GPT-2 [156], trained on an unlabeled dataset of 8 million web pages. Data collection for training emphasizes content variety while neglecting ideological balance, prompting various critical inquiries:

> *Does the current crop of large-scale generative language models serve to further entrench ideologically biased political beliefs? Is it possible to steer them towards creating information that is not biased by political ideology?*

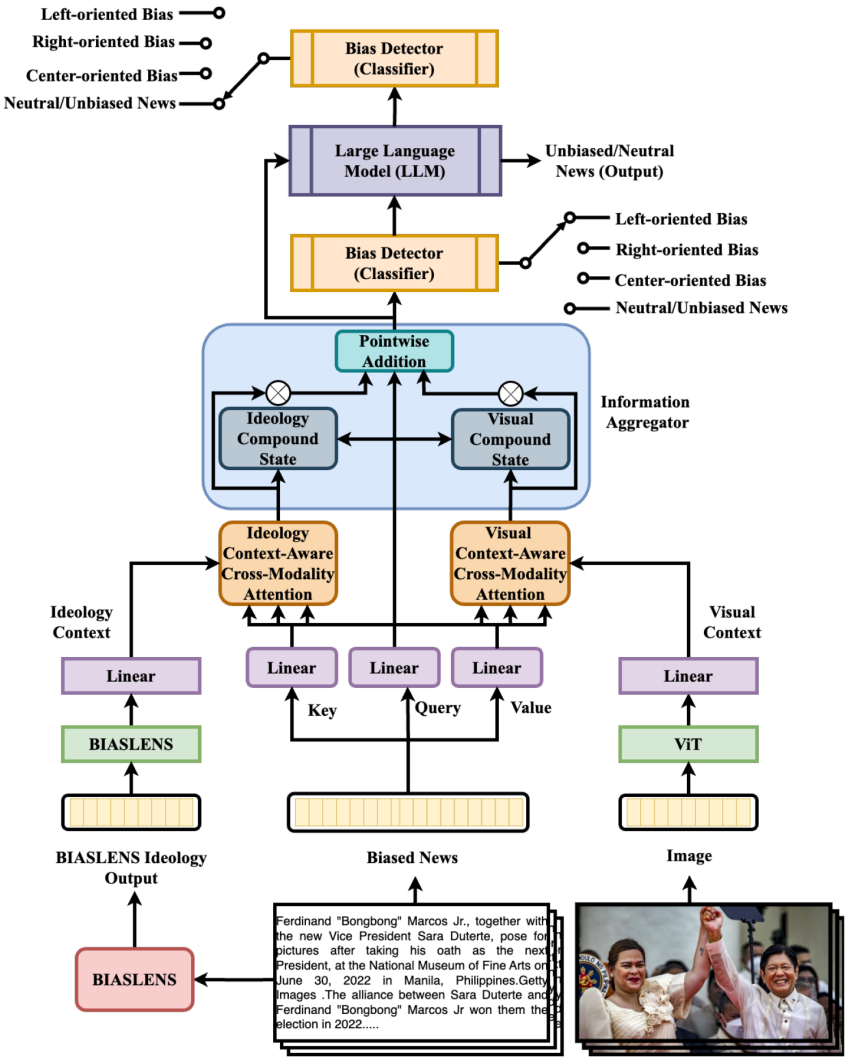


Fig. 9. Architecture of CANDOR (ideological polarity debiaser).

The latest achievements in neural machine translation and language modeling using transformer-based encoder–decoder architectures had a substantial impact on the design of our media bias neutralizer. As illustrated in Figure 9, we initially modify the encoder's input to the summation of the *ideology-information infused* and *visual-information infused* embeddings, denoted as H_i and H_v, respectively. This is referred to as the *multimodal information fused representation*, $\hat{H}$, specifically the output of the *Information Aggregator* block.

4.3.1 Language Modeling. A series of predefined tokens is often used to initiate auto-regressive LLMs [156]. In this case, an LLM will generate a sequence of T tokens when given a prompt ψ. We have $X = [x_t]$, while $t \in [1, T]$ in which the variable x_t is defined:

$$x_t \sim \underset{\hat{x_t}}{argmax} P(\hat{x_t}) = LLM(x_{1:t-1}|\psi). \tag{10}$$

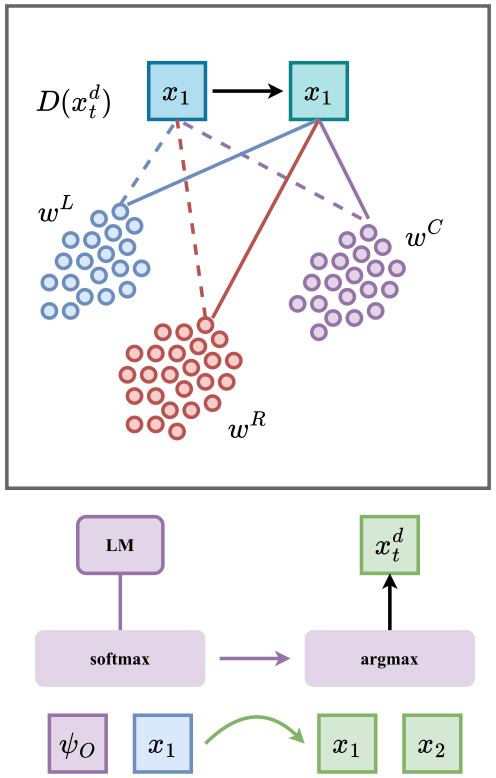


Fig. 10. Lexical neutralization framework.

4.3.2 Progressive Bias Mitigation. We maintain the primary architecture of the LLM model while adjusting the bias during generation, in contrast to the approach that integrates fairness loss and retrains an unbiased LLM from the ground up. Figure 10 illustrates the incorporation of a debiasing step, using word embeddings, between the softmax and argmax functions. A number of rounds of reinforced optimization are used to calibrate the standard generation in order to get tokens that are not biased.

The *state* is defined at step t in reinforcement learning as $s_t = x_{1:t}$, which includes all sequences created prior to t, and the action at step t is $a_t = x_t$, which is the t-th output token. Given the state $s_t = x_{1:t}$, it indicates the probability of selecting token x_t (action a_t), the softmax output of the last hidden states is used as the policy π_θ. We further compile a predetermined set of politically biased terms: w^L for left-leaning bias, w^C for center-leaning bias, and w^R for right-leaning bias, which are derived from the ALLSIDES and BIGNEWS datasets via TF-IDF.

4.3.3 Fairness Incentive Mechanism. The reward function is pivotal in steering the optimization process toward generating unbiased content while preserving factual accuracy and coherence. In our framework, we employ a *contrastive reward function*, which effectively balances bias mitigation with content preservation. This function quantitatively assesses the divergence between the generated output and the original biased text, ensuring the retention of essential information. This approach addresses the challenge of media bias mitigation, where the objective is to neutralize polarizing language without compromising the journalistic integrity of the source material.

Formally, the fairness incentive function is defined as:

$$R(x_t^d) = \frac{\pi_{\theta_d}(a_t|s_t)}{\pi_\theta(a_t|s_t) + \epsilon} - \gamma \left\| x_t^d - x_t^b \right\|_2 + \lambda D(x_t^d), \tag{11}$$

where the first term, referred to as *Policy Ratio Term* quantifies the divergence between the debiased policy π_{θ_d} and the original biased policy π_θ, encouraging a controlled shift toward neutrality. A higher ratio reflects effective bias reduction while maintaining relevance. However, excessive divergence may compromise factual accuracy, which is undesirable in journalistic contexts. The denominator includes a small constant ϵ to prevent instability and ensure robustness during optimization. The second term, *Content Preservation Penalty*, introduces L^2 norm penalty which ensures the debiased output x_t^d remains semantically and syntactically similar to the original biased content x_t^b. This penalty acts as a content-preserving constraint, preventing the model from producing outputs that are overly restructured or semantically distant from the source. The coefficient λ controls content retention, balancing neutrality with factual accuracy. A higher γ enforces stricter content preservation, reducing the risk of factual distortion. The final term, $\lambda D(x_t^d)$, known as *Bias Distance Penalty*, represents a bias quantification function that measures the ideological inclination of the debiased output. It penalizes residual bias in the generated content, ensuring that the model actively minimizes politically charged language and ideological framing. The term $D(x_t^d)$ represents the debias gain derived from *Lexical Neutralization Framework* (Section 4.3.4), which serves as a signal to direct the debias generation. This term reinforces neutrality by discouraging the presence of ideologically loaded phrases, thus enhancing objectivity in the output. The parameter λ controls the influence of the bias penalty, balancing fluency and neutrality. Bias quantification is performed using an embedding-based metric that evaluates the ideological proximity of the generated text to known biased language patterns.

The contrastive reward function thus effectively balances bias mitigation with content preservation, enabling the model to produce ideologically neutral yet factually consistent text. This approach aligns with existing work on bias reduction in language models, such as Park et al. [146] and Dong et al. [47], who demonstrated the effectiveness of contrastive learning in mitigating bias across diverse applications.

4.3.4 Lexical Neutralization Framework. In the task of ideological bias neutralization, ensuring that the generated content remains factually accurate while being free from inherent bias is paramount. The *Lexical Neutralization Framework* serves as a critical component of our research, leveraging carefully designed distance-based functions to quantify and minimize bias at the lexical level. By leveraging *weighted debias gain* and *entropy-based embedding* calculations, the framework ensures the generation of semantically neutral text while maintaining syntactic fluency and content preservation. Specifically, the framework operates by analyzing the proximity of generated text to ideologically charged terms from left-leaning, right-leaning, and center-leaning sources. This section details the formulation and technical components of our debiasing strategy.

According to research in the field of unbiased word embedding, one proven technique is to keep neutral words evenly spaced from groups of sensitive words inside the embedding space [237]. Rather than using it as an objective to develop unbiased LLMs, we use it as a criterion to choose the unbiased token at each stage of creation. This component aims to adjust the semantic positioning of generated tokens by enforcing a separation margin from ideologically charged terms while aligning them with more moderate terms. Specifically, considering the lists of *left-leaning*, *right-leaning*, and *center-leaning* words denoted as w^L, w^C, and w^R, respectively, the *weighted debias gain* $D(x_t^d)$ for

the token x_t^d is:

$$D(x_t^d) = \sum_{w \in w^L \cup w^R} \Big(max(0, \delta - dist(x_t^d, w) + \alpha \cdot dist(x_t^d, w^C)) \Big). \tag{12}$$

Here, $w^L \cup w^R$ represents the union of left-leaning and right-leaning word lists. The summation ensures comprehensive bias mitigation by considering the entire set of biased terms. The distance function $dist(x_t^d, w^C)$ measure the *entropy-based embedding* distance between the generated token x_t^d and biased center-bias word list, w^C. Here, δ is a hyperparameter defining the separation margin that establishes the minimum acceptable distance from biased terms, and α represents a weighting factor that determines the relative importance of alignment with neutral terms (w^C). A value of $0 < \alpha < 1$ assigns relative importance to moderate terms to prevent over reliance on them. Overall, it encourages controlled semantic drift toward neutrality without distorting factual content. In Equation (12), the first term, $max(0, \delta - dist(x_t^d, w)$, penalizes the generated token x_t^d if it is within a threshold (δ) of ideologically charged terms (w^L or w^R). This encourages the system to push the token away from biased terms. The second term, $\alpha \cdot dist(x_t^d, w^C)$, incentivizes alignment with center-leaning (neutral) terms (w^C). The weight parameter α controls the strength of this alignment, balancing neutrality with semantic relevance. This component acts as a guardrail to ensure that the generated text moves away from polarized language while maintaining a degree of alignment with moderate terms.

While the *weight debias gain* addresses the positional alignment of individual tokens, the *entropy-based embedding* distance calculation ensures a balanced distribution of probabilities across the ideologically charged and neutral terms. This metric is mathematically expressed as:

$$dist(x_t^d, w) = - \sum_{w \in w^L \cup w^R} P(w|x_t^d) \log P(w|x_t^d) - \alpha \sum_{w\prime \in w^C} P(w\prime|x_t^d) \log P(w\prime|x_t^d). \tag{13}$$

Here, $P(w|x_t^d) \log P(w|x_t^d)$ is the probability of generating an ideologically charged term (w^L or w^R) conditioned on the generated token x_t^d, $P(w\prime|x_t^d) \log P(w\prime|x_t^d)$ represents the probability of generating a center-neutral term (w^C) conditioned on x_t^d, and lastly α is a weighting factor that determines the relative importance of neutral terms in the probability distribution. The first term, $- \sum_{w \in w^L \cup w^R} P(w|x_t^d) \log P(w|x_t^d)$, measures the entropy of the probabilities associated with left- and right-leaning terms. Lower entropy here indicates a skewed probability distribution, which is penalized to ensure ideological neutrality. The second term, $\alpha \sum_{w\prime \in w^C} P(w\prime|x_t^d) \log P(w\prime|x_t^d)$, promotes a balanced alignment with neutral terms. The parameter α adjusts the importance of this term, ensuring the generated content is not overly biased while retaining factual integrity. All together, it promotes diversity in word selection, reducing residual bias.

Together, Equations (12) and (13) guide the model to generate tokens that minimize proximity to ideologically charged language while maintaining a balance between neutrality and fluency. The *weighted bias adjustment* ensures that individual tokens are repositioned in the semantic space to avoid bias, while the *entropy-based embedding distance* calculation reinforces this neutrality over a sequence of generated tokens. Consider a news article discussing a controversial policy decision. The original biased text might include terms such as "radical socialist policies" (left-oriented bias) or "extreme capitalist agenda" (right-oriented bias). The *lexical neutralization framework*: (1) penalizes the inclusion of terms like "radical" and "extreme" based on their proximity to ideologically charged terms in the embedding space; (2) encourages the generation of neutral alternatives, such as "ambitious policies" or "market-oriented agenda"; (3) balances the probability distribution of generated terms to ensure semantic coherence and neutrality.

Algorithm 1: Persistent Political Impartiality (Contrastive Reward)

Input: Word lists with bias w^L, w^R, and w^C, KL-divergence threshold σ, content preservation weight γ
for $t = 1, 2, \ldots$ **do**
  Produce $(a_t|s_t)$ as trajectories using the vanilla policy π_θ; Compute $D(x_t^d)$ (Equation 12);
  Estimate reward $R(x_t^d)$ with:

$$R(x_t^d) = \frac{\pi_{\theta_d}(a_t|s_t)}{\pi_\theta(a_t|s_t) + \epsilon} - \gamma \left\| x_t^d - x_t^b \right\|_2 + \lambda D(x_t^d) \tag{14}$$

  Compute policy update:

$$\theta_d \leftarrow \arg\max_\theta \lambda_t R(x_t^d)(\theta) - KL(\theta||\theta_d) \tag{15}$$

  by using Adaptive Momentum (Adam) to take K steps of SGD; **if** $KL(\theta||\theta_d) \geq 2\sigma$ **then**
    $\lambda_{t+1} = \lambda_t/2$;
  else
    if $KL(\theta||\theta_d) \leq \sigma/2$ **then**
      $\lambda_{t+1} = 2\lambda_t$;
    end
  end
end
Give back the impartial policy π_{θ_d};

4.3.5 Adaptive Bias Adjustment. In addition to the fairness incentive, we further include as an auxiliary constraint the **Kullback-Leibler (KL)** divergence between the original distribution θ and the debiased θ_d to avoid a debias policy that heavily deviates from the original policy, potentially resulting in reduced readability. Algorithm 1 displays the methods we used for bias calibration.

Based on the KL divergence of the current step, to adaptively modify the debiasing (fairness incentive) and semantic coherence (KL constraint) strengths, we set the balance parameter λ_t and target divergence σ. The debias technique is termed "calibration" since it does not create unbiased text from the outset, but instead mitigates bias in the hidden states (with parameter θ) of standard creation. The algorithm generates a debiased policy π_{θ_d} that enables the production of text according to political neutrality.

4.4 Vision-Text Context Transformer (VT-CT)

A self-attention network may directly link to another long-range representation by computing attention weights for every word pair inside a single sequence. In self-attention networks, contextual information is disregarded in the computation of relationships among the constituents. Evidence from studies mentioned in [206, 214] suggests that attention models, in particular, might benefit from including contextual information when modeling relationships among neural representations. The conventional dot-product cross-modal attention method facilitates direct interaction between textual representations and other modalities. Since each modality originates from a distinct embedding subspace, directly fusing multimodal information may not preserve optimal contextual information and might introduce significant noise into the final representations. Informed by the research of Yang et al. [225], we advocate for multimodal fusion via *Context-Aware Cross-Modality Attention*. Initially, we produce multimodal information-conditioned key and value vectors, followed by the execution of conventional scaled dot-product attention.

4.4.1 Motivation. We propose to enhance self-attention networks by using contextual richness. We eliminate reliance on external resources, such as the embeddings of previous sentences, by using internal representations as context vectors. In particular, we contextualize the updates made by the input, query, and key layers into context so that they may calculate the items' relative

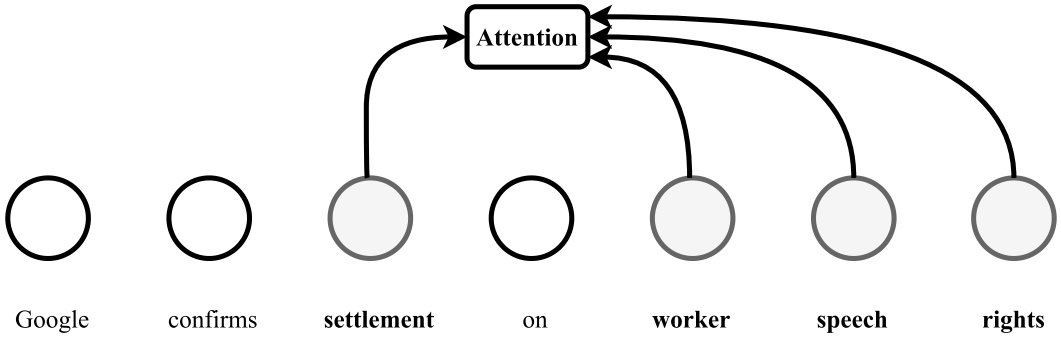


Fig. 11. Conventional self-attention network.

importance:

$$QK^T = (HW_Q)(HW_K)^T = H(W_Q W_K)^T H^T. \tag{16}$$

The importance between word pairs is computed via the self-attention model, as shown in Figure 11 ("settlement," "worker," "speech," and "rights") independently, disregarding contextual information.

4.4.2 Approach. To address the issue of missing contextual information and to keep self-attention networks' parallel computation flexibility intact, we suggest contextualizing the changes from input layer H to the query and key layers. We specifically adhere to Shaw et al. [175] to transmit contextual information to transformation via addition, hence minimizing computational escalation.

We compute the query, key, and value vectors from the intermediate representation H produced by **Generative PLMs (GPLMs)** at a designated layer. Let $C \in \mathbb{R}^{n \times d_{context}}$ represent the vector derived from the visual representation. We provide multimodal information, namely informed key and value vectors $\hat{K}$ and $\hat{V}$, as delineated by Yang et al. [225]. To determine the quantity of information to preserve from the textual modality, we derive the matrix $\lambda \in \mathbb{R}^{n \times 1}$ (Equation (18)). Observe that U_k and $U_v \in \mathbb{R}^{d_{context} \times d}$ are matrices subject to learning:

$$\begin{bmatrix} \hat{K} \\ \hat{V} \end{bmatrix} = \left(I - \begin{bmatrix} \lambda_k \\ \lambda_v \end{bmatrix} \right) + \begin{bmatrix} \lambda_k \\ \lambda_v \end{bmatrix} \left(C \begin{bmatrix} U_k \\ U_v \end{bmatrix} \right). \tag{17}$$

According to Britz et al. [29] and Vaswani et al. [210], regions with very small gradients may be driven into by a large magnitude of Q and K in the softmax function. To mitigate this impact, $\{\lambda_k, \lambda_v\}$ may be used as a factor to modulate the size of $\hat{K}$ and $\hat{V}$. Drawing upon previous research on multimodal networks [227], we allow the model to determine the values of λ_k and λ_v via a gating mechanism, rather than treating them as hyper-parameters, as outlined in Equation (18):

$$\begin{bmatrix} \lambda_k \\ \lambda_v \end{bmatrix} = \sigma \left(\begin{bmatrix} K \\ V \end{bmatrix} \begin{bmatrix} W_{k_1} \\ W_{k_2} \end{bmatrix} \right) + C \left(\begin{bmatrix} U_k \\ U_v \end{bmatrix} \begin{bmatrix} W_{k_2} \\ W_{v_2} \end{bmatrix} \right). \tag{18}$$

The matrices $W_{k_1}, W_{k_2}, W_{v_1},$ and $W_{v_2} \in \mathbb{R}^{d \times 1}$ are trained concurrently with the model, and $\sigma(.)$ represents the logistic sigmoid function. By using the gating scalar, the model is able to determine with remarkable precision the relative importance of each representation and context vector in predicting attention weight.

Ultimately, the multimodal information-augmented vectors $\hat{K}$ and $\hat{V}$ are used to calculate the conventional scaled dot-product attention. In our scenario, we possess two supplementary modalities: image and ideology. Employing the context-aware cross-modality attention mechanism, we derive the *ideology-information infused* and *visual-information infused* vectors H_i and H_v, respectively (Equations (19) and (20)):

$$H_i = softmax\left(\frac{Q \cdot \hat{K}_i^T}{\sqrt{d_k}}\right)\hat{V}_i, \tag{19}$$

$$H_v = softmax\left(\frac{Q \cdot \hat{K}_v^T}{\sqrt{d_k}}\right)\hat{V}_v. \tag{20}$$

4.4.3 Choices of Context Vectors. Maintaining the concept of self-attention's adaptability and simplicity is a key tenet of our approach. Therefore, we eliminate reliance on external resources by using the internal states as context vectors. We utilize a wide variety of context vectors, any of which may be used alone or in composite form.

Unified Multimodal Context. The overall significance of a sequence is represented by the input layer, which in turn determines the unified multimodal context. We encapsulate the input layer representations using the mean, a method often utilized in Seq2Seq models [36]:

$$C = \bar{H} \in \mathbb{R}^d. \tag{21}$$

The unified multimodal context, which is distributed over layer states, is a vector as opposed to a matrix. The self-attention model's unified multimodal context is considered an instance-specific bias that aims to improve the common unified parameters $\{W_Q, W_K\}$ in the training dataset. An instance-specific prior is produced by combining pairwise characteristics with global features; its effectiveness in different detection and identification tasks is shown by its use in different research conducted by Hariharan et al. [77] and Zhu et al. [238].

Layered Multimodal Context. The layers that constitute the layered multimodal context are those that are located below the current input layer. Advanced neural models are often constructed with numerous layers that can capture various forms of syntactic and semantic information [147]. Peters et al. [147] show that word meaning is influenced by context and that higher-level states include these aspects, whereas lower-level states represent syntactic elements, and concurrently revealing all these signals is significantly advantageous. In this case, we will pretend that H^l at the lth level is the current input layer and that the collection of layers below it is the layered multimodal context:

$$C = [H^1, ..., H^{l-1}] \in \mathbb{R}^{n \times (l-1)d}. \tag{22}$$

With the use of layered multimodal context, the self-attention model may include a wide variety of syntactic and semantic data from different levels.

Instead of using a linear combination, we use a dense connection strategy. Our technique diverges from that of Peters et al. [147] in the following aspects: The use of positive and negative parameters gives us more leeway in the models we develop, as opposed to their normalized weights. Secondly, they apply a scalar uniformly across all items in the layer states, whereas for every element, we allocate a distinct "scalar." According to Tu et al. [206], the second option allows for more control over the combination since it allows the model to be more expressive compared to scalars.

Hierarchical Fusion Context. By combining unified multimodal context across layers, we may naturally combine the concepts of deep and unified multimodal context:

$$C = [c^1, ..., c^l] \in \mathbb{R}^{ld}, \tag{23}$$

where c^l is the unified multimodal context of the lth layer H^l, computed according to Equation (21). We anticipate that the deep-global environment may provide varying degrees of language biases, including lexical, syntactic, and semantic levels.

4.5 Information Aggregator

To integrate information from both ideological and visual modalities, we develop the *Information Aggregator* block. We suggest two gates: the *ideology gate* (g_i) and the *visual gate* (g_v), to regulate the volume of information sent by each modality. The following are:

$$g_i = [H \oplus H_i]W_i + b_i, \tag{24}$$

$$g_v = [H \oplus H_v]W_v + b_v. \tag{25}$$

In this context, $W_i, W_v \in \mathbb{R}^{2d \times d}$ and $b_i, b_v \in \mathbb{R}^{d \times 1}$ are trainable parameters, whereas $\oplus$ signifies concatenation. The ultimate *multimodal information fused representation* $\hat{H}$ is specified by Equation (26):

$$\hat{H} = H + g_i \odot H_i + g_v \odot H_v. \tag{26}$$

The vector $\hat{H}$ is inserted back into the GPLM for further processing.

5 Experimental Setup

To put our system into action, we train a generative LLM, a BIASLENS model to extract ideology traits, and a media polarity detector.

5.1 Data Pre-Processing

We take the data and pre-process it before we enter it into our model. A substantial quantity of media-related stuff is included in the dataset (such as HTML elements, ads, etc.). Since our data is collected straight from published publications. We eliminate any extraneous text and retain just the article's core section. All extraneous punctuation marks have been eliminated. The information about paragraphs, including the markers for new lines, is kept as it might be useful for our project.

5.2 Models

We selected the pre-existing LLaMA-2, GPT-2, and FLAN-T5 as the generative LLMs for our research. The RoBERTa-based domain-adaptive pre-trained polarity detector was trained on the WNC, with parameters fine-tuned to achieve optimal performance (F1 = 0.886). We also evaluated BERT [45], ELECTRA [38], DistilBERT [167], and DistilRoBERTa; among these, the domain-adaptive pre-trained RoBERTa exhibited superior performance. Likewise, BIASLENS is pre-trained using the BIGNEWS dataset.

5.3 Training Detail

The ALLSIDES, WNC, and BIGNEWS datasets were used individually for different pre-training, fine-tuning and testing tasks where all datasets are split into three sets: a training set comprising 70% of the total, a validation set accounting for 10% of the total, and a test set including 20% of the whole. The stratified K-fold cross-validation method ($K = 5$) has also been employed to maintain the statistical consistency of category representation, ensuring robust and fair model evaluation. This strategy addresses overfitting concerns with smaller K values and variance issues with larger K

values. Each dataset was divided into five equally stratified folds, enabling each sample to be used for both training and validation. In addition to the standard training-validation-testing protocol, this cross-validation framework was adopted to assess model stability and reduce the bias introduced by any single train-test split. The average performance metrics and standard deviations across these 5 folds are reported in *Supplementary Materials; Section D*; *Table 4*.

During training, we used the early halting strategy when no progress was noticed. We trained our media bias neutralizer model with two NVIDIA Tesla V100 GPUs, each equipped with 32 GB of VRAM. A default value of 0.6 is initially assigned to the debias strength parameter λ, and its performance is evaluated at $\lambda = \{0.1, 0.3, 0.5, 0.7, 0.9\}$. We selected around 400 bias words corresponding to one of the three polarities: left-oriented, right-oriented, and center-oriented. The KL divergence threshold σ is established at 0.02.

6 Evaluation

Utilizing previous research, we use several automated criteria to assess our model's efficacy in bias mitigation, information retention, and fluency. The following section elucidates each assessment, with the comprehensive results shown in Table 11. Example generations are shown in *Supplementary Materials; Section C*, *Table 3*. We assess the outputs from three models: the standard GPT-2 (baseline), a lexical neutralization approach combined with the BIASLENS model for ideological prediction, all integrated inside a context-aware cross-modality attention and gated information aggregator.

6.1 VT-CT

In this part, we performed experiments to assess the influence of various context vector types on the news debiasing tasks using a Transformer-based model. We started by looking into how different context vectors affected the self-attention networks on the encoder side. Our next step was to find out whether the decoder-side self-attention network consistently improved when we included contextual information. In the end, we tested how well the context-aware model might perform with self-attention networks on the encoder and decoder sides. We treat the choice of context vector structure and injection location (encoder, decoder, both) as architectural hyperparameters and analyze their impact. In order to minimize the impact of the control variable, we restricted the first two ablation experiments to self-attention networks on the encoder or decoder sides. This analysis provides architectural guidance on selecting context representation and injection strategy to balance performance (BLEU) with model complexity (parameters).

6.1.1 Encoder. Table 6 demonstrates that all suggested context vector techniques consistently enhance model performance compared to the baseline, therefore, the significance of incorporating contextual information into self-attention networks is affirmed. *Model #2* (*unified multimodal context*) and *Model #3* (*layered multimodal context*) both accomplished comparable advancements. Different levels of global linguistic biases improve translation effectiveness, as shown by the fact that hierarchical fusion context is superior to its global counterpart. Furthermore, we assessed whether the global and deep approaches are complementary. The model "*hierarchical fusion context + layered multimodal context* (*Model #5*)" accomplishes further improvements simply by combining them into the final context vectors. The results show that self-attention networks are improved in several ways by using the two types of context vectors.

6.1.2 Decoder. In this series of tests, we sought to determine which context vectors were best for use in decoder-side self-attention networks. According to Table 6, the self-attention networks are consistently improved by both "*hierarchical fusion context + layered multimodal context*" (*Model #7*). Nonetheless, the latter surpasses the former, mirroring the phenomena seen in the trials with the encoder. Zhang et al. [232] observed that the self-attention networks on the decoder

Table 6. Experimental Results on the Selection of "Context Vectors" for the Task of News Debiasing

Sl. No. (#)	Model	Applied to	Context Vectors	# Param	BLEU
1	BASE	n/a	n/a	88M	27.31
2	VT-CT (Proposed)	Encoder	Unified Multimodal Context	93M	27.55
3			Layered Multimodal Context	97.2M	28.02
4			Hierarchical Fusion Context	104.6M	28.56
5			Layered Multimodal Context + Hierarchical Fusion Context	116M	**29.53**
6		Decoder	Hierarchical Fusion Context	104.6M	28.94
7			Layered Multimodal Context + Hierarchical Fusion Context	116M	29.25
8		Encoder + Decoder	Layered Multimodal Context + Hierarchical Fusion Context	132.8M	29.43

A model's trainable parameter size is represented by "Param," where "M" = Million. The highest values are shown by **bold** scores.

side concentrate only on adjacent representations. However, our findings show that decoder-side self-attention networks rely on both the forward (global) and bottom (deep) representations.

6.1.3 Encoder and Decoder. In the end, the encoder and decoder were both enhanced using these concepts. The neutralized news quality is substantially diminished by this technique (*Model #8*) as compared to encoder-side models. The usual encoder–decoder attention paradigm in Transformers is believed to be responsible for the BLEU score decline since it depends on the top layer of encoder representation—which inherently contains relevant contextual information—in its operations. The Transformer architecture's encoder-side self-attention network may provide more benefits to the context-aware model.

6.2 BIASLENS

In light of the significance of ideology/philosophy prediction tasks in political science, we perform comprehensive experiments over a diverse array of datasets including six tasks. Subsequently, we conduct a comparison with both traditional models and previous PLMs (Section 6.2.1), as well as among our model variations (Section 6.2.2). In Section 6.2.3, we provide and analyze results, demonstrating that BIASLENS surpasses all baseline models in 5 out of 6 tasks. All models are trained using a subset of the BIGNEWS dataset, referred to as BIGNEWSALIGN, for ideological and narrative aims.

6.2.1 Baselines. We take a look at three baselines. At first, a linear SVM is trained using unigram and bigram features, since this is considered the typical baseline in political science (Yu 2008 Classifying, Diermeier 2012 Language). We also use the standard fine-tuning technique for the ideology prediction task to compare with BERT and RoBERTa.

6.2.2 Model Variants. We examine many iterations of BIASLENS. Initially, we conduct experiments on models trained only with the triplet loss objective, using either the ideology/philosophy objective, the narrative objective, or a combination of both.

6.2.3 Main Results. All tasks' F1 scores are shown in Table 7. BIASLENS achieves a higher overall average F1 score than the most robust baseline, RoBERTa, by a margin of 5.5%. Significantly, BIASLENS surpasses all three baselines stated in Section 6.2.1 on 5 out of 6 activities, achieving an

Table 7. F1 Scores on Six Evaluation Tasks

	Ideology Prediction					
	Congressional Speech	**Hyperpartisan**	**ALLSIDES**	**YouTube Cmt**	**YouTube User**	**Twitter**
Baselines						
SVM	71.31	61.25	52.51	65.34	66.49	42.85
BERT	65.88	48.42	60.88	64.64	65.24	44.20
RoBERTa	67.25	65.43	69.75	64.72	71.98	47.90
Baly et al. [18]						
With Original Data	66.74	58.37	72.89	65.42	70.47	44.95
With BIGNEWSBLN	70.39	71.24	<u>76.47</u>	<u>68.57</u>	76.47	74.74
Using the triplet loss objective only						
Ideology Objective	72.12	<u>71.58</u>	**76.70**	67.23	78.07	44.45
Story Objective	<u>72.89</u>	69.34	74.93	67.56	<u>78.89</u>	<u>48.67</u>
Ideology Objective + Story Objective (BIASLENS) (Proposed)	**73.05**	**72.24**	76.23	**68.65**	**79.32**	**50.56**

The findings marked as "best" are in bold, while those marked as "second best" are underlined.

improvement above 10.20% in ideology labeling when used on YouTube user levels and Hyperpartisan. We attribute the increase in performance to the ideology-driven pre-training target that we proposed, which successfully captures political content. It is noteworthy that, in some activities, other model versions surpass BIASLENS by a little margin, which may be relevant to practitioners engaged in certain jobs.

The approach presented by Baly et al. [18] is similar in that it utilizes triplet loss as a pre-training objective, but it is limited to articles dealing with the same topics. Based on the dataset that was originally published by Baly et al. [18] and BIGNEWSBLN, we run two versions of their model. As far as ideological prediction tasks are concerned, pre-training on our BIGNEWSBLN seems to outperform using the actual data, highlighting BIGNEWSBLN's importance. Secondly, the triplet construction technique proposed by Baly et al. [18] using BIGNEWSBLN exhibits poor generalization in the ideology prediction task when compared to BIASLENS and its variations. In light of this, it is clear that our goals of making it easier to compare articles from different sources on the same topics are beneficial. Our ideology-driven aims facilitate the acquisition of information necessary to recognize ideology.

6.2.4 Comparison with Previous State-of-the-Arts. According to Table 8, BIASLENS outperforms previous state-of-the-art results by 5.5 points, achieving an accuracy of 86.1%, when applied to hyperpartisan data using the *original* binary prediction framework (i.e., hyperpartisan or not) [98].

6.2.5 On Texts of Different Characteristics. Our investigation into the model's performance is extended to data that exhibits varying degrees of *formality in the language*, *training size*, *document length*, and *aggregation level*, as shown in Table 5. BIASLENS demonstrates enhanced efficacy on more formal and extended texts, since its pre-training is conducted on news items. BIASLENS demonstrates resilience with training datasets of limited size.

Table 8. Evaluation in Relation to the Prior State-of-the-Art Hyperpartisan Models That Used the Initial Binary Prediction Configuration

Models	Hyperpartisan			
	Accuracy	**Precision**	**Recall**	**F1 Score**
ELMO + CNN [93]	82.2	87.1	75.5	80.9
Logistic Regression [191]	82.0	81.5	82.8	82.1
BERT [76]	80.9	82.3	78.7	80.5
SVM [88]	80.6	85.8	73.2	79.0
ULMFiT [231]	80.3	79.3	81.8	80.6
RoBERTa	84.3	87.2	80.6	83.7
BIASLENS	**86.1**	**87.5**	**84.8**	**86.1**

Bold are the best results. The highest levels of accuracy, precision, recall, and F1 are achieved with BIASLENS.

Table 9. Ablation Study Results on BIASLENS

	Ideology Prediction					
	Congressional Speech	**Hyperpartisan**	**ALLSIDES**	**YouTube Cmt**	**YouTube User**	**Twitter**
BIASLENS	73.05	72.24	76.23	68.65	79.32	50.56
No Ideology Objective	−1.23	−14.53	−2.43	−3.41	−11.65	−4.35
No Story Objective	+1.44	−0.98	+1.72	**+2.78**	+1.76	−2.86

Best results are in bold.

6.2.6 Ablation Study. We demonstrate the effect of eliminating each ideology-driven pretraining target from BIASLENS in Table 9. Initially, the elimination of the ideological objective yields the greatest damage. This further illustrates the efficacy of our triplet-loss concept about identical narrative pieces. Eliminating the narrative objective lowers overall performance by around 1%, while somewhat enhancing ideology prediction. This indicates that the narrative objective functions as further limitations to prevent over-fitting on the "shortcuts" for identifying ideologies.

6.2.7 Effect of Varying β in Loss Weighting. We treat β as a key hyperparameter in the joint loss formulation and evaluate its influence across datasets to guide optimal configuration. We perform a focused ablation study to assess the impact of the tradeoff parameter β, given in Equation (9), which weights the ideology loss $\mathcal{L}_{ideology}$ versus the narrative loss $\mathcal{L}_{story}$. We evaluate β in $\{0.0, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ across three representative datasets: Congressional Speech, Hyperpartisan News, and YouTube User Comments. As shown in Table 10, moderate weighting of ideology ($\beta = 0.2$) consistently yields the best tradeoff between generalizability and ideological discrimination. When $\beta = 1.0$, the model overfits on ideology "shortcuts," degrading performance on out-of-domain data (e.g., YouTube). Conversely, at $\beta = 0.0$, the model weakens in ideological identification despite narrative robustness.

Table 10. Performance Comparison for Different β Values on Representative Datasets

β	Congressional Speech (F1)	Hyperpartisan (F1)	YouTube User (F1)
0.0	70.94 (2.11 ↓)	68.45 (3.79 ↓)	**79.78** (0.46 ↑)
0.1	72.23 (0.82 ↓)	71.43 (0.81 ↓)	78.26 (1.06 ↓)
0.2 (Chosen)	**73.05**	**72.24**	**79.32**
0.4	71.56 (1.49 ↓)	70.68 (1.56 ↓)	78.89 (0.43 ↓)
0.6	70.87 (2.18 ↓)	69.42 (2.82 ↓)	77.85 (1.47 ↓)
0.8	69.44 (3.61 ↓)	68.79 (3.45 ↓)	76.71 (2.61 ↓)
1.0	67.33 (5.72 ↓)	65.21 (7.03 ↓)	72.36 (6.96 ↓)

The highest values are shown by **bold** scores. The values indicated by the downward (↓) and upward (↑) arrows denote the absolute change (decrease/increase) in F1 score, illustrating how the performance on different datasets—Congressional Speech, Hyperpartisan, and YouTube User—varies with different values of β tested during the ablation study, relative to the reference setting β=2.

Table 11. Critical Analysis of Our Approach in Light of Previous Best-in-Class Alternatives (Pryzant et al. [154] on Bias Neutralization, Mitra et al. [138] on Domain-Adaptive Bias Mitigation Network) and Other State-of-the-Art Models Based on Style Transfer and DRL [94, 113, 151, 219, 235]

Evaluation Outcome on ALLSIDES Dataset with CANDOR Framework, with $\lambda = 0.7$											
Mode	**Type**	**Models**	**Left-oriented (L) → Neutral (N)**								
			Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**	**LexBiasRed**
Headlines	Baselines (Vanilla Model)	BART	0.5213	0.6234	0.5677	103.12	0.5034	0.2843	0.4556	0.5524	0.5012
		T5	0.5253	0.6467	0.5797	101.56	0.5968	0.2943	0.4628	0.6827	0.5185
		GPT-2	0.6345	0.7089	0.6696	101.89	0.6264	0.3244	0.4796	0.7653	0.5120
	Pryzant et al. [154]		0.6674	0.6821	0.6746	103.75	0.6235	0.3345	0.4535	0.7353	0.4910
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.5678	0.6257	0.5953	102.57	0.5454	0.3101	0.4989	0.5916	0.5323
		DABDMN-T5	0.6652	0.6867	0.6757	102.88	0.6454	0.3489	0.5033	0.7116	0.5423
		DABDMN-GPT2	0.6898	0.7012	0.6954	101.67	0.7068	0.3639	0.5110	0.7997	0.5512
	DRG [113]		0.7046	0.6857	0.6950	102.68	0.6735	0.3378	0.5212	0.6568	0.5173
	BT [151]		0.7097	0.6984	0.7049	103.52	0.6787	0.3556	0.5345	0.6725	0.5316
	DRL [94]		0.7214	0.7019	0.7115	102.87	0.6967	0.3856	0.5648	0.7001	0.5395
	Style Transfer [235]		0.7174	0.6935	0.7052	102.75	0.6864	0.3814	0.5562	0.6996	0.5507
	PTO [219]		0.7125	0.6884	0.7002	103.87	0.6676	0.3647	0.4948	0.6843	0.5393
	CANDOR (Proposed)	GPT-2	0.7354	**0.7897**	0.7615	102.34	0.6402	0.5487	0.6843	0.7932	0.5995
		FLAN-T5	0.7232	0.7234	0.7232	103.45	0.6845	0.4882	0.5412	0.7909	0.5832
		LLaMA-2	**0.7567**	0.7708	**0.7636**	**100.72**	**0.7143**	**0.5647**	**0.6892**	**0.8022**	**0.6016**
Text Body	Baselines (Vanilla Model)	BART	0.6879	0.7453	0.7154	115.23	0.5134	0.2945	0.4642	0.5657	0.4988
		T5	0.6374	0.7234	0.6776	113.56	0.5845	0.2856	0.4542	0.6745	0.5010
		GPT-2	0.6784	0.7867	0.7285	114.58	0.6312	0.3243	0.4698	0.7635	0.5037
	Pryzant et al. [154]		0.8016	0.7975	0.7995	114.36	0.5876	0.3013	0.4332	0.7124	0.4929
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.6873	0.7230	0.7046	111.43	0.5523	0.3312	0.4978	0.6132	0.5330
		DABDMN-T5	0.7685	0.7978	0.7828	110.89	0.6534	0.3465	0.5134	0.7013	0.5477
		DABDMN-GPT2	0.8452	0.8052	0.8247	112.54	0.6897	0.3675	0.5345	0.7789	0.5618
	DRG [113]		0.8474	0.8257	0.8364	113.83	0.6612	0.3857	0.5535	0.7263	0.5009
	BT [151]		0.8523	0.8230	0.8373	113.75	0.6625	0.3936	0.5682	0.7334	0.5200
	DRL [94]		0.8636	0.8336	0.8483	111.97	0.6735	0.4073	0.5710	0.7461	0.5300
	Style Transfer [235]		0.8668	0.8423	0.8543	111.10	0.6773	0.4257	0.5846	0.7485	0.5423
	PTO [219]		0.8453	0.8252	0.8351	114.04	0.6575	0.3845	0.5523	0.7276	0.5209
	CANDOR (Proposed)	GPT-2	0.8856	0.8456	0.8651	109.65	0.6543	0.5534	**0.6576**	0.7687	0.6026
		FLAN-T5	0.8252	0.8572	0.8408	112.46	0.6456	0.5123	0.6243	0.7532	0.6021
		LLaMA - 2	**0.8907**	**0.8961**	**0.8933**	**109.57**	**0.6980**	**0.5657**	0.6546	**0.7809**	**0.6043**
Mode	**Type**	**Models**	**Right-oriented (R) → Neutral (N)**								
			Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**	**LexBiasRed**
Headlines	Baselines (Vanilla Model)	BART	0.5575	0.6453	0.5981	108.31	0.5032	0.2879	0.4365	0.5587	0.4978
		T5	0.6454	0.6978	0.6705	106.62	0.5780	0.2869	0.4576	0.6786	0.5099
		GPT-2	0.6367	0.7234	0.6772	106.84	0.6245	0.3345	0.4765	0.7698	0.4994
	Pryzant et al. [154]		0.7107	0.6849	0.6975	107.35	0.6234	0.3445	0.4545	0.7324	0.4902
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.6467	0.6886	0.6669	107.17	0.5654	0.3134	0.4567	0.5867	0.5382
		DABDMN-T5	0.6572	0.7245	0.6896	106.85	0.6543	0.3423	0.5123	0.7275	0.5428
		DABDMN-GPT2	0.7436	0.7342	0.7388	107.03	0.7032	0.3675	0.5312	**0.7879**	**0.5964**
	DRG [113]		0.6654	0.7023	0.6833	107.34	0.6668	0.3721	0.5526	0.7367	0.5155

(Continued)

Table 11. Continued

Evaluation Outcome on ALLSIDES Dataset with CANDOR Framework, with $\lambda = 0.7$										
	BT [151]		0.6772	0.7058	0.6912	107.23	0.6678	0.3835	0.5623	0.7403	0.5338
	DRL [94]		0.6951	0.7145	0.7046	106.43	0.6775	0.3887	0.5678	0.7531	0.5424
	Style Transfer [235]		0.7055	0.7157	0.7105	106.14	0.6835	0.4034	0.5832	0.7652	0.5502
	PTO [219]		0.6731	0.6925	0.6826	107.52	0.6645	0.3743	0.5567	0.7351	0.5347
	CANDOR (Proposed)	GPT-2	0.7089	0.7534	0.7304	**105.30**	0.6465	0.5465	**0.6781**	0.7789	0.5868
		FLAN-T5	0.7254	0.7023	0.7136	106.53	0.6789	0.4953	0.5465	0.7798	0.5829
		LLaMA-2	**0.7897**	**0.7653**	**0.7773**	105.43	**0.7132**	**0.5654**	0.6645	0.7778	0.5961
Text Body	Baselines (Vanilla Model)	BART	0.6576	0.7311	0.6924	113.62	0.5123	0.2980	0.4435	0.5645	0.4942
		T5	0.7342	0.7125	0.7231	112.87	0.5867	0.2908	0.4546	0.6547	0.5141
		GPT-2	0.7365	0.7897	0.7621	112.66	0.6564	0.3340	0.4786	0.7791	0.4922
	Pryzant et al. [154]		0.7807	0.7651	0.7728	113.65	0.6234	0.3015	0.4129	0.7431	0.4949
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.6789	0.7012	0.6898	112.53	0.5456	0.3123	0.4890	0.6032	0.5416
		DABDMN-T5	0.7655	0.7789	0.7721	112.34	0.6430	0.3421	0.5239	0.7275	0.5430
		DABDMN-GPT2	0.8625	0.8673	0.8648	111.41	0.6567	0.3456	0.5364	0.7687	0.5501
	DRG [113]		0.7714	0.7843	0.7777	112.66	0.6528	0.3406	0.5254	0.7452	0.5163
	BT [151]		0.7834	0.7923	0.7878	112.64	0.6541	0.3574	0.5302	0.7521	0.5294
	DRL [94]		0.8250	0.8174	0.8211	111.75	0.6615	0.3758	0.5541	0.7597	0.5377
	Style Transfer [235]		0.8310	0.8263	0.8286	111.58	0.6613	0.3682	0.5567	0.7583	0.5500
	PTO [219]		0.7856	0.7843	0.7849	112.02	0.6456	0.3447	0.5347	0.7521	0.5336
	CANDOR (Proposed)	GPT-2	0.8644	0.8753	0.8698	111.25	0.6654	0.5543	**0.6867**	**0.7871**	0.5817
		FLAN-T5	0.8257	**0.8825**	0.8531	112.50	0.6576	0.5342	0.6534	0.7645	0.5922
		LLaMA - 2	**0.8974**	0.8745	**0.8858**	**110.23**	**0.6867**	**0.5678**	0.6745	0.7768	**0.6166**
Mode	**Type**	**Models**	**Center-oriented (C) → Neutral (N)**								
			Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**	**LexBiasRed**
Headlines	Baselines (Vanilla Model)	BART	0.5234	0.6345	0.5736	118.42	0.5031	0.2987	0.4645	0.5765	0.5166
		T5	0.6365	0.6892	0.6618	118.92	0.5786	0.2897	0.4476	0.6789	0.5042
		GPT-2	0.6789	0.7545	0.7147	117.63	0.6134	0.3432	0.4567	0.7671	0.4936
	Pryzant et al. [154]		0.7118	0.7377	0.7245	116.14	0.6343	0.3342	0.4325	0.7112	0.4921
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.6682	0.6869	0.6774	118.47	0.5543	0.3143	0.4234	0.5786	0.5414
		DABDMN-T5	0.6761	0.7462	0.7094	115.53	0.6456	0.3312	0.5129	0.7234	0.5484
		DABDMN-GPT2	0.7456	0.7873	0.7658	115.74	0.7042	0.3867	0.5346	0.7789	0.5654
	DRG [113]		0.6846	0.7525	0.7169	117.56	0.6376	0.3453	0.5285	0.7362	0.5099
	BT [151]		0.7004	0.7457	0.7223	116.87	0.6460	0.3556	0.5324	0.7470	0.5240
	DRL [94]		0.7270	0.7435	0.7351	116.60	0.6521	0.3664	0.5561	0.7479	0.5366
	Style Transfer [235]		0.7367	0.7523	0.7444	115.56	0.6568	0.3865	0.5473	0.7547	0.5438
	PTO [219]		0.6946	0.7447	0.7187	117.61	0.6427	0.3469	0.5278	0.7421	0.5331
	CANDOR (Proposed)	GPT-2	0.7576	0.7967	0.7766	115.48	0.6675	**0.5765**	0.6643	0.7897	0.5895
		FLAN-T5	0.7445	0.7523	0.7483	115.34	0.6785	0.4765	0.5567	0.7923	0.5856
		LLaMA-2	**0.8089**	**0.8243**	**0.8165**	**114.83**	**0.7134**	0.5576	**0.6788**	**0.8134**	**0.6029**
Text Body	Baselines (Vanilla Model)	BART	0.6734	0.6677	0.6705	128.25	0.5032	0.2900	0.4576	0.5674	0.4987
		T5	0.6232	0.7421	0.6774	126.94	0.5978	0.2812	0.4687	0.6567	0.4948
		GPT-2	0.7134	0.7532	0.7327	126.25	0.6534	0.3564	0.4743	0.7678	0.5179
	Pryzant et al. [154]		0.7506	0.7460	0.7482	126.56	0.5657	0.3230	0.4149	0.7414	0.4931
	Domain-Adaptive Bias Detection and Mitigation Network (DABDMN) [138]	DABDMN-BART	0.6897	0.6978	0.6937	126.34	0.5546	0.3342	0.4435	0.5923	0.5395
		DABDMN-T5	0.6780	0.7445	0.7096	125.63	0.6405	0.3345	0.5324	0.7145	0.5531
		DABDMN-GPT2	0.7456	0.7890	0.7666	125.42	0.6809	0.3567	0.5546	0.7456	0.5661
	DRG [113]		0.6952	0.7534	0.7231	126.36	0.6526	0.3568	0.5412	0.7277	0.5037
	BT [151]		0.7023	0.7596	0.7298	126.12	0.6576	0.3667	0.5583	0.7351	0.5329
	DRL [94]		0.7141	0.7842	0.7475	125.89	0.6673	0.3864	0.5631	0.7476	0.5358
	Style Transfer [235]		0.7251	0.7946	0.7582	124.76	0.6746	0.3923	0.5821	0.7536	0.5511
	PTO [219]		0.6823	0.7135	0.6975	126.31	0.6487	0.3472	0.5350	0.7371	0.5279
	CANDOR (Proposed)	GPT-2	0.7890	0.8346	0.8111	125.40	0.6687	**0.5709**	0.6721	0.7789	0.5914
		FLAN-T5	0.7778	**0.8634**	0.8183	124.54	0.6729	0.5564	0.6543	0.7678	0.5817
		LLaMA - 2	**0.8434**	0.8308	**0.8370**	**122.32**	**0.6908**	0.5675	**0.6869**	**0.7890**	**0.6007**

BT, Back-Translation; DRG, Delete-Retrieve-Generate; DRL, Disentangled Representation Learning; PTO, Point-then-Operate; UST, Unsupervised Machine Translation.
These experiments demonstrate principled hyperparameter tuning to achieve an optimal tradeoff between ideological discrimination and generalization across domains.

These experiments demonstrate principled hyperparameter tuning to achieve an optimal tradeoff between ideological discrimination and generalization across domains.

6.3 Media Bias Neutralizer (Ideological Polarity Debiaser)

Using our dataset, we train the DABDMN-RoBERTa [138] polarity classifier to serve as an independent, external evaluation of ideological polarity debiasing. The classifier was trained for 200 epochs and contains three classes: left-oriented, center-oriented, and right-oriented. It also uses n-grams up to three. On the validation dataset, the classifier achieves an F1 score of 0.886. The generated sentences' polarity labels are anticipated using this classifier. The success of our bias

mitigation is measured by how often the generated words fool this classifier into classifying them with the desired bias-neutralized label. The percentage of neutralized words that are labeled with the neutral label is called the debiasing accuracy, or s_p. The findings shown in Table 11 indicate that our technique surpasses the previous work [154] by a minimum of 10–13% in the context of headlines, achieved by neutralizing all news articles. The previous research by Liu et al. [122] employs complete body flipping and is suitable for comparison. Nonetheless, it is evident that the model's effectiveness diminishes while analyzing entire body text compared to headlines. A potential reason is the absence of robust structural parallelism between the headlines and the body content. Our model's capacity to learn bias-invariant information throughout the whole text is hindered by this.

Perhaps we might test the efficacy of our polarity detector by doing ablation studies. According to Table 11, the detector greatly improves the neutralization accuracy. This is logical, since the polarity detector directs the model's attention to statements that explicitly exhibit bias. Along with the comparison based on the training-validation-test set, we also report the mean and standard deviation of precision, recall, and F1-score across five stratified folds on the ALLSIDES dataset for different competing models. As shown in *Supplementary Materials; Section D*; *Table 4*, the results demonstrate consistent performance, validating the robustness of our model across different data.

6.3.1 Content Preservation. We compare the generated phrases to the original ones using the measure developed by Fu et al. [54], where the similarity of sentences is determined by embedding. In both the source and target phrases, for every token, we first generate embedding vectors using a pre-trained DABDMN-RoBERTa model. After that, we take the word embeddings inside a phrase and use *max*, *min*, and *mean pooling* to determine the sentence embeddings. With the use of cosine similarity based on embeddings, we can find out how similar the produced sentences are to the originals in terms of meaning:

$$s_c = \sum_{i=1}^{D_{val}} cos(v^-, v^+). \tag{27}$$

From the test set, we create an inverted text (v^-) and compare it to the original text (v^+) to determine their semantic similarity.

The polarity debiasing and content maintaining scores are often contradictory. A superior mitigating score often results in a diminished content preservation score, and conversely. Consequently, we amalgamate these ratings into a singular metric referred to as the "Overall" score.

6.3.2 Overall. The media bias neutralizer may get great accuracy effortlessly if content preservation is not required. Likewise, it may achieve flawless content retention by directly replicating the original input into the inverted creation. In order to measure how well our framework works, we use the harmonic mean of the polarity debiasing performance metrics (s_p) and content preservation metrics (s_c), which captures this tradeoff relationship in the evaluations:

$$s_o = \frac{2 s_c s_p}{s_c + s_p}. \tag{28}$$

Table 11 demonstrates that our model surpasses previous efforts in the total score for both headline and news text debiasing challenges. According to the ablation analysis of the body text, adding the polarity detector improves the overall score.

6.3.3 Fluency. **Perplexity (PPL)** is utilized as an indicator of fluency. In the context of a LLM-estimated sequence probability $p(w_1, w_2, ..., w_M)$ and a sequence of M words $w_1, w_2, ..., w_M$, the

following definition of PPL applies:

$$PPL = p(w_1, w_2, ..., w_M)^{-\frac{1}{M}}. \tag{29}$$

One popular way to measure a language model's accuracy in predicting a given sentence is by looking at its PPL. In other words, if an LLM is trained on an extensive English corpus, it can assess the degree to which a particular phrase conforms to the English language, reflecting its fluency. To train a language model on the ALLSIDES dataset, we use kenLM [80] due to its efficiency and speed. We use this language model to assess the ambiguity of created phrases. Reduced ambiguity ratings indicate enhanced fluency in created sentences. Table 11 displays the outcomes. Compared to headlines, created body paragraphs are generally more confusing. This shows that digesting lengthier content is more challenging for the model. As part of the headline bias-neutralization challenge, our approach does not provide a significant benefit over previous techniques regarding confusion. We change the order of thirty percent of the characters in the input text.

6.3.4 Extended Baseline Comparisons with Style Transfer and Disentangled Models. We provide a more rigorous and nuanced evaluation of the generative component in CANDOR by extending our baseline comparisons. Models specifically designed for attribute-controlled text rewriting, ideological or sentiment transformation tasks are served as comparative baselines to benchmark CANDOR's ability to neutralize ideological bias while preserving semantic integrity and linguistic fluency.

The **Delete-Retrieve-Generate (DRG)** model [113] introduces an interpretable three-step pipeline: it identifies and removes attribute-specific tokens (*Delete*), retrieves a candidate sentence from the target domain with similar content (*Retrieve*), and finally generates the neutralized output (*Generate*) using a decoder conditioned on the retrieved target. Despite its clarity and controllability, DRG lacks end-to-end adaptability for ideology-aware transformations and performs sub-optimally in cross-domain political bias mitigation. The **Back-Translation (BT)** framework [151] exploits machine translation as a bridge for style transfer. The input is translated to a pivot language (e.g., French), and then back to English using a decoder trained on the target style. **Disentangled Representation Learning (DRL)**, [94] learns separate latent variables for style and content via a multiobjective autoencoder framework. It minimizes reconstruction loss $\mathcal{L}_{rec}$, style prediction loss $\mathcal{L}_{style}$, and adversarial losses for content invariance. The latent representation z is decomposed as:

$$z = [z_c; z_s], \quad \text{where } z_c \perp z_s. \tag{30}$$

This separation is enforced via orthogonality constraints and mutual information minimization between z_c (content) and z_s (style). While theoretically elegant, this model suffers from instability during training and fails to generalize to multimodal or low-resource bias scenarios.

The *Style Transfer as* **Unsupervised Machine Translation (UST)** model [235] treats style transfer as unsupervised translation using denoising autoencoders and BT, for minimizing:

$$\mathcal{L}_{UST} = \mathcal{L}_{recon} + \mathcal{L}_{cycle} + \mathcal{L}_{adv}, \tag{31}$$

where $\mathcal{L}_{cycle}$ ensures cycle consistency, and $\mathcal{L}_{adv}$ encourages stylistic indistinguishability. This model provides strong rewriting fluency but lacks interpretability and performs poorly on ideologically entangled news inputs without explicit polarity control.

Lastly, the **Point-Then-Operate (PTO)** framework proposed by We et al. [219] introduces a hierarchical pipeline that first detects stylistic edit points using a pointing mechanism and then applies template-aware rewrite operations. The sequence generation is performed in two stages:

$$Pointer : x \rightarrow p_i; \quad Operator : (x, p_i) \rightarrow y. \tag{32}$$

This division improves control and alignment but requires fine-grained annotations and performs inconsistently under sparse ideology labels or ambiguous input structures.

In Table 11, we compare our CANDOR framework against these baselines, using identical metrics across bias-neutralization accuracy (s_p), content preservation (s_c), overall harmonic score (s_o), PPL, and lexical overlap (ROUGE/BLEU). We retrain and evaluate all models on the ALLSIDES benchmark using identical train/text splits. Our model consistently outperforms these baselines due to its modular architecture, ideology-aware contextual encoding and reinforcement-based neutrality objection. Furthermore, unlike the aforementioned methods, CANDOR seamlessly integrates multimodal context vectors and supports policy-aligned generation, making it better suited for real-world ideological debiasing tasks in both headlines and full-body political discourse.

6.3.5 Tradeoff between Debias and Fluency. We treat λ as a critical hyperparameter in the reward function of the CANDOR bias neutralization module and analyze its tradeoffs across fluency, bias removal, and content preservation. Our early research showed that debiased outputs with a greater debias strength parameter ($\lambda \rightarrow 1$) occasionally had more syntactic errors, suggesting that the model aggressively attempts to eliminate bias while sacrificing some semantic fluency. The tradeoff between generative fluency and bias reduction is examined in this section. In order to evaluate PPL, we train three separate language models on the standard generation for our three features using kenLM [80]. Table 12 illustrates that, generally, models trained with bigger λ produce texts with increased PPL (decreased fluency) so corroborating our first discovery. The complexity remains quite low, even when using the maximal debias strength parameter of $\lambda = 0.7$, across all three features. This might be attributed, in part, to the adaptive handling of the KL constraint in Algorithm 1. This λ-based hyperparameter analysis ensures that the selected configuration yields optimal rewriting fidelity under both automated and human assessments.

6.3.6 Ablation Study. The ablation study (Table 13) evaluates the effectiveness of various components of the proposed CANDOR framework for news bias neutralization. The purpose of the research is to evaluate the impact of the CANDOR framework's individual parts, including:

(1) The significance of visual cues in bias detection and neutralization for better generalization.
(2) The *BIASLENS* module (ideology-aware bias detection).
(3) The *context vector mechanisms* derived from the VT-CT.

By systematically removing these components, the ablation study evaluates their individual and combined impact on various performance metrics:

Analysis with Unimodal Textual Data. We introduce two explicit configurations: *CANDOR* (*Headlines Only*) and *CANDOR* (*Text Body Only*) which quantify the standalone contributions of short-form and long-form text inputs, respectively. The results, as reflected in Table 13, show a consistent drop in performance metrics, including accuracy, BLEU, and content preservation when the visual context is excluded or input richness is reduced. This substantiates the hypothesis that multimodal context, particularly images, enhances bias detection and neutralization in news narratives by capturing complementary cues not present in text alone.

Impact of Removing BIASLENS. Removing BIASLENS leads to a noticeable drop in accuracy and content preservation, highlighting its crucial role in detecting ideological bias and guiding neutralization. For example, in the headline task, accuracy drops significantly without BIASLENS.

Impact of Removing Context Vectors. The absence of context vectors results in performance reduction in accuracy and overall content preservation. This demonstrates the importance of multimodal context (text and visual) in enhancing bias detection and text generation.

Table 12. Tradeoff between Accuracy, Content Preservation, Overall, and PPL

Tradeoff among Accuracy, Content, Overall, and PPL for CANDOR Framework with LLaMA-2 LLM							
		Left-oriented (L) → Neutral (N)					
Mode	λ	$\lambda = 0$ **(reference)**	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
Headlines	Accuracy	0.6536	0.03 ↑	0.05 ↑	0.06 ↑	0.10 ↑	0.02 ↓
	Content	0.6891	0.02 ↑	0.03 ↑	0.06 ↑	0.08 ↑	0.01 ↓
	Overall	0.6708	0.01 ↑	0.04 ↑	0.07 ↑	0.09 ↑	0.10 ↑
	PPL	103.45	103.23	102.47	101.31	100.72	103.76
Text Body	Accuracy	0.8032	0.02 ↑	0.05 ↑	0.04 ↑	0.08 ↑	0.09 ↑
	Content	0.8246	0.01 ↑	0.05 ↑	0.06 ↑	0.07 ↑	0.12 ↓
	Overall	0.8137	0.02 ↑	0.06 ↑	0.08 ↑	0.07 ↑	0.01 ↓
	PPL	112.46	112.39	111.53	110.51	109.57	110.46
		Right-oriented (R) → Neutral (N)					
Mode	λ	$\lambda = 0$ **(reference)**	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
Headlines	Accuracy	0.6735	0.03 ↑	0.04 ↑	0.07 ↑	0.11 ↑	0.02 ↓
	Content	0.6929	0.01 ↑	0.05 ↑	0.03 ↑	0.07 ↑	0.01 ↓
	Overall	0.6830	0.03 ↑	0.05 ↑	0.06 ↑	0.09 ↑	0.20 ↓
	PPL	107.75	107.23	106.51	106.66	105.30	105.68
Text Body	Accuracy	0.8063	0.03 ↑	0.05 ↑	0.08 ↑	0.09 ↑	0.15 ↑
	Content	0.8248	0.01 ↑	0.03 ↑	0.05 ↑	0.05 ↑	0.06 ↑
	Overall	0.8154	0.02 ↑	0.03 ↑	0.06 ↑	0.07 ↑	0.10 ↓
	PPL	112.63	113.69	112.31	111.68	110.23	111.76
		Center-oriented (C) → Neutral (N)					
Mode	λ	$\lambda = 0$ **(reference)**	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
Headlines	Accuracy	0.7435	0.01 ↑	0.04 ↑	0.05 ↑	0.06 ↑	0.10 ↓
	Content	0.7878	0.01 ↑	0.02 ↑	0.027 ↑	0.03 ↑	0.04 ↑
	Overall	0.7650	0.02 ↑	0.03 ↑	0.039 ↑	0.05 ↑	0.02 ↓
	PPL	117.46	117.12	166.52	115.74	114.83	115.76
Text Body	Accuracy	0.7835	0.02 ↑	0.03 ↑	0.06 ↑	0.05 ↓	0.01 ↑
	Content	0.7943	0.01 ↑	0.02 ↑	0.03 ↑	0.03 ↑	0.04 ↓
	Overall	0.7888	0.01 ↑	0.03 ↑	0.03 ↑	0.04 ↑	0.10 ↓
	PPL	126.25	125.46	125.12	124.81	122.32	123.41

The debias strength parameter λ begins at 0 (no debias, vanilla production) and rises to 0.9 over time. ↑ and ↓ indicate changes compared with $\lambda = 0$.

It is important to distinguish between the CANDOR (Headlines Only) or CANDOR (Text Body Only) modes and the CANDOR (w/o Context Vector) ablation. The former evaluates the model's performance using only textual inputs, either headlines or full news bodies without any image-based features, thereby isolating the contribution of visual signals (refer to *Supplementary Materials; Section E*; *Figure 2*). In contrast, the *w/o Context Vector* configuration still leverages both textual and visual inputs, but disables the multimodal context fusion mechanism. A precise interpretation of the ablation results reported in Table 13.

Combined Framework (Full CANDOR). The complete CANDOR framework, incorporating BIASLENS and VT-CT, achieves the highest scores across all metrics. It excels particularly in

Table 13. Ablation Study Performed on CANDOR Framework with LLaMA-2 Based LLM, Tested upon ALLSIDES Dataset, with the Debias Strength ($\lambda = 0.7$)

Ablation Study on LLaMA-2-based CANDOR Framework, with $\lambda = 0.7$									
Mode	**Type**	**Left-oriented (L) → Neutral (N)**							
		Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**
Headlines	CANDOR (Headlines Only)	0.6856	0.6989	0.6921	102.35	0.6167	0.5023	0.6247	0.7412
	CANDOR (w/o BIASLENS)	0.7076	0.7001	0.7038	103.64	0.6512	0.5146	0.6242	0.7564
	CANDOR (w/o Context Vector)	0.7345	0.7452	0.7398	102.45	0.6867	0.5346	0.6501	0.7811
	CANDOR	**0.7567**	**0.7708**	**0.7636**	**100.72**	**0.7143**	**0.5647**	**0.6892**	**0.8022**
Text Body	CANDOR (Text Body Only)	0.8042	0.8294	0.8166	114.25	0.6493	0.5331	0.6086	0.6281
	CANDOR (w/o BIASLENS)	0.8455	0.8457	0.8455	111.42	0.6557	0.5358	0.6123	0.6468
	CANDOR (w/o Context Vector)	0.8674	0.8646	0.8812	110.47	0.6756	0.5413	0.6286	0.6743
	CANDOR	**0.8907**	**0.8961**	**0.8933**	**109.57**	**0.6980**	**0.5657**	**0.6546**	**0.7089**
Mode	**Type**	**Right-oriented (R) → Neutral (N)**							
		Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**
Headlines	CANDOR (Headlines Only)	0.7334	0.7363	0.7348	106.31	0.6673	0.5256	0.6487	0.7325
	CANDOR (w/o BIASLENS)	0.7586	0.7462	0.7523	105.72	0.6823	0.5377	0.6501	0.7468
	CANDOR (w/o Context Vector)	0.7744	0.7523	0.7631	105.36	0.6962	0.5471	0.6576	0.7573
	CANDOR	**0.7897**	**0.7653**	**0.7773**	**105.30**	**0.7132**	**0.5654**	**0.6645**	**0.7778**
Text Body	CANDOR (Text Body Only)	0.8456	0.8374	0.8414	113.64	0.6361	0.5378	0.6357	0.7310
	CANDOR (w/o BIASLENS)	0.8567	0.8423	0.8494	112.47	0.6437	0.5474	0.6591	0.7598
	CANDOR (w/o Context Vector)	0.8746	0.8645	0.8695	110.65	0.6578	0.5523	0.6612	0.7510
	CANDOR	**0.8974**	**0.8745**	**0.8858**	**110.23**	**0.6867**	**0.5678**	**0.6745**	**0.7768**
Mode	**Type**	**Center-oriented (C) → Neutral (N)**							
		Acc.	**Cont.**	**Over.**	**PPL**	**R-1**	**R-2**	**R-L**	**BLEU**
Headlines	CANDOR (Headlines Only)	0.7418	0.7606	0.7510	116.21	0.6341	0.5026	0.5957	0.7695
	CANDOR (w/o BIASLENS)	0.7524	0.7612	0.7567	115.58	0.6468	0.5176	0.6157	0.7957
	CANDOR (w/o Context Vector)	0.7736	0.7851	0.7793	115.04	0.6645	0.5232	0.6235	0.7857
	CANDOR	**0.8089**	**0.8243**	**0.8165**	**114.83**	**0.7134**	**0.5576**	**0.6788**	**0.8134**
Text Body	CANDOR (Text Body Only)	0.7569	0.7617	0.7592	128.53	0.6271	0.5263	0.6226	0.7287
	CANDOR (w/o BIASLENS)	0.7834	0.7823	0.7828	125.11	0.6476	0.5317	0.6445	0.7424
	CANDOR (w/o Context Vector)	0.8023	0.7923	0.7972	124.57	0.6527	0.5487	0.6523	0.7540
	CANDOR	**0.8434**	**0.8308**	**0.8370**	**122.32**	**0.6908**	**0.5675**	**0.6869**	**0.7890**

The highest values are shown by **bold** scores.

PPL, ROUGE, and BLEU, indicating its ability to generate fluent, semantically accurate, and unbiased text.

The ablation study conclusively shows that both *BIASLENS* and the *context vectors* from VT-CT are indispensable for attaining state-of-the-art performance in news polarity detection and bias neutralization.

6.4 Lexical Bias Quantification and Comparative Interpretability

While CANDOR's evaluation pipeline emphasizes semantic preservation (via semantic similarity scores), polarity reduction (via bias classifier shifts), and fluency (via PPL), these metrics, though robust, are not always intuitively interpretable to human evaluators. We incorporate *lexical bias quantification*, a concept-driven approach grounded in ideologically marked tokens and biased framing patterns.

6.4.1 Lexical Bias Scoring Framework. We define a *lexical bias score* for a sentence $x = (w_1, w_2, \ldots, w_n)$ as:

$$s_{\text{bias}}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\text{bias}}(w_i), \tag{33}$$

where $\mathbf{1}_{\text{bias}}(w_i) = 1$ if token w_i appears in a curated or automatically mined bias lexicon, and 0 otherwise. We utilize the *BABE dataset* [188] and *Media Bias Lexicon Toolkit* [155, 181], to identify such tokens. These include assertives (e.g., "clearly," "undeniable"), presuppositions (e.g., "even," "again"), and framing cues (e.g., "radical," "elites"). We computed token-level lexical bias scores for each such phrase and assigned a numerical polarity score.

6.4.2 Comparative Analysis with CANDOR. To evaluate alignment with semantic debiasing, we compute $s_{\text{bias}}(x)$ for each original input sentence and its CANDOR-generated neutralized counterpart. We then report:

(1) *Lexical Bias Reduction* $\Delta s_{\text{bias}} = s_{\text{bias}}(x_{\text{input}}) - s_{\text{bias}}(x_{\text{CANDOR}})$.
(2) *Correlation* with polarity score change Δs_p (from our bias classifier).
(3) *Correlation* with content preservation score s_c.

Empirical results show that CANDOR's neutralization consistently reduces lexical bias by an average of 58.4% across left, right, and center-leaning headlines. However, some cases with *high semantic preservation* exhibit *residual lexical bias*, indicating divergence. Conversely, cases with *complete lexical sanitization* sometimes show *low semantic similarity*, suggesting overcorrection or semantic drift. Table 11 provides a comparative analysis of lexical bias reduction across various stylistic transfer models and CANDOR variants. CANDOR-LLaMA2 achieved the highest lexical bias reduction, indicating string performance not just in semantic and classifier-based polarity reduction but also in symbolic bias terms. Other stylistic transfer models like BT [151] and DRG [113] exhibited comparatively lower lexical sanitization.

We also compare with *concept-based ideology embeddings*, such as the method of ideology-anchored projections used by Baly et al. [18]. Outputs of CANDOR showed notable realignment toward the neutral centroid in this embedding space, indicating ideological balancing from a concept geometry perspective. Additionally, we did a comparative study using the Gaussian bias distribution model by Chen et al. [33], which measures bias based on the frequency and positional distributions of ideological sentences. Results show CANDOR's outputs fall within the low-bias Gaussian cluster of neutral texts, in contrast to baseline style-transfer models.

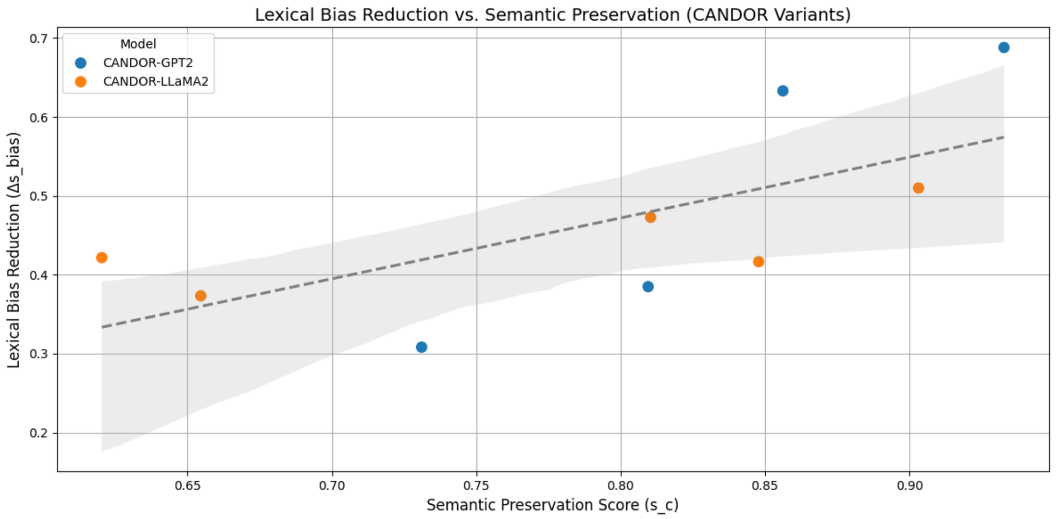


Fig. 12. Tradeoff and alignment between lexical santization and semantic fidelity.

6.4.3 Interpretability Benefit. Lexical bias scoring offers a surface-level, human-intuitive signal of ideological tone, which when cross-referenced with classifier-based polarity shifts s_p and semantic preservation s_c, creates a *triangulated framework* for evaluation. This triangulation with symbolic, conceptual, and learned-space bias estimators demonstrate that CANDOR's debiasing is not only perceived by humans but also measurable via interpretable ideologically-aware signals. Aligning with recent work on faithful political stance estimation [155, 181], Figure 12 visualizes the tradeoff and alignment between lexical sanitization and semantic fidelity, emphasizing the dual necessity of symbolic and contextual fairness in debiasing. A positive trend is visible in case models preserving content meaning better also tend to clean lexical bias more effectively. However, the dashed regression line shows that perfect alignment is not guaranteed, supporting the point that semantic and lexical indicators must be jointly evaluated for holistic debiasing assessment.

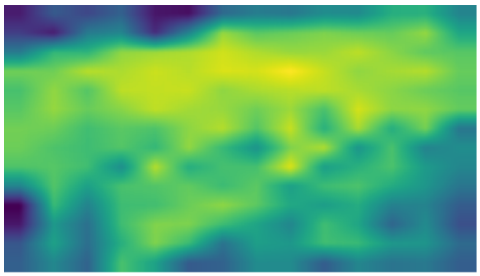
We plan to extend this lexical quantification with *rhetorical role detection* and *framing analysis* as proposed by Spinde et al. [181, 184].

6.5 Multimodal Attribution: The Role of Visual Content in Debiasing

CANDOR's architecture includes a vision-language fusion pipeline that integrates contextual signals from accompanying visual content (e.g., news images, memes, political illustrations). Beyond serving as an auxiliary context, we explore how visual cues directly influence ideological debiasing and generation.

Visual Feature Attribution. We adopt a gradient-based attribution framework adapted from CLIP GradCAM [105, 236, 240], and integrated it with our VT-CT fusion mechanism. For each input image-text pair (x^v, x^t), we compute attention-weighted saliency maps over image regions that most influence (a) the polarity classifier's decision, and (b) the decoder's style control head during rewriting. Results (Figure 13 and *Supplementary Materials; Section F, Figure 3*) show that ideologically symbolic regions, such as political party symbols, flags, and so on. receive high attribution weights. In particular, attention visualizations from VT-CT confirm that visual embeddings guide the model to focus on emotionally charged or partisan tokens

(a) The original input image associated with a right-leaning news source (Fox News), covering a politically charged event.



(b) The attention heatmap generated using CLIP-based GradCAM and a Gaussian smoothing filter, highlighting high-attribution regions learned by the model.

(c) The same attention scores overlayed on the original image to provide visual interpretability.

Fig. 13. Visual attribution maps reveal ideological cues in multimodal debiasing. The highlighted regions (e.g., "TRUMP 2020" signage, red hats, crowd patterns) indicate that the model identifies visually salient and ideologically symbolic cues. These cues directly influence polarity classification and style-conditioned rewriting, reinforcing that the visual stream plays a non-trivial role in contextual disambiguation and debiasing.

(e.g, "illegals," "radical left," "fake news"), leading to more robust neutralization in ideologically polarized settings.

Ablation Study. To isolate the contribution of visual signals, we compare the rewriting quality of the full multimodal model against a text-only ablation (where the image branch is masked). As shown in Table 13, the multimodal model exhibits higher bias reduction (up to +5.1%) and lower PPL in polarized contexts involving charged imagery. Particularly in cases where the textual tone is ambiguous, visual grounding disambiguates ideological framing, improving the accuracy and neutrality of rewriting.

6.6 Ethical Implications and Societal Risks

Here, CANDOR framework presents a complex landscape of ethical and societal considerations too. This concern becomes especially apparent when analyzing real-world headlines. For instance, a conservative-leaning headline such as "*Climate Alarmism Hurts Working Families, Experts Say*" and a progressive-leaning one like "*Oil Lobbyists Hijack Climate Policy to Delay Action*" both represent distinct and valid ideological perspectives. If both were neutralized into a generalized statement such as "*Experts Weigh Impacts of Climate Policy Discussions*," the resulting abstraction may obscure crucial ideological framing and weaken the communicative intent of the original message.

The CANDOR framework incorporates a series of architectural and algorithmic safeguards, designed to preserve content fidelity while minimizing ideological bias. The BIASLENS module encourages positioning the texts within an embedding space, preventing naive or one-dimensional rewriting of partisan content. Additionally, the reinforcement learning-based debiasing module is governed by a composite reward function (Equation (11)) that simultaneously optimizes for bias reduction, content preservation, and linguistic fluency.

Despite these safeguards, it is imperative to acknowledge that systems like CANDOR may still be susceptible to subtle forms of epistemic bias, particularly when trained on datasets with imbalanced ideological representation. To minimize these risks, we advocate for the inclusion of metadata in deployment environments that clearly indicates when and how content has been algorithmically modified. Furthermore, we recommend that model outputs be logged and subjected to periodic fairness audits to ensure that the system does not disproportionately alter narratives from specific ideological categories. In parallel, alignment with regulatory frameworks ensures transparency and human agency in decision-making processes.

7 Human Evaluation

By means of a web-based experiment conducted on Amazon Mechanical Turk (MTurk), we evaluated how well our method altered human readers' perceptions of the polarity of news pieces. MTurk is a website that allows users to collect data via crowdsourcing, including a diverse workforce [30].

7.1 Experimental Design

Using MTurk, we recruited 378 (75.6%) male and 122 (24.4%) female US participants ($N = 500$), with minimum age 18 and average age close to 36 years. Participants must possess an approval rate over 95% and have an average period of education of 14 years. Participants are asked for basic information like their age, gender, level of education, preferred media, political affiliation, and on which part of the political spectrum they stood (*1—Very liberal*, *4—Moderate*, *7—Very conservative*). Upon self-reporting their party identification, 190 participants identified as Democrats (38%), 230 as Republicans (46%), and 80 as independents (16%).

For evaluation we used 10 sets of headlines as a test and 15 sets of full articles (body text), with five sets representing right-leaning, five sets representing center-leaning, and five sets representing left-leaning ideologies. Each work consists of two versions: original and neutral. Every set in the neutralization task has one left-oriented, center-oriented, or right-oriented article or headline that was originally biased, and our framework-neutralized one article or headline in each set. The participants were randomly given the task of reading either two sets of headlines or two sets of body content.

7.1.1 Procedure. Participants are instructed to read and evaluate each version (original and neutralized) independently within each set based on readability, perceived bias (or political polarity), and corresponding political ideology. Participants are required to evaluate whether the version exhibits more impartiality, as well as to determine if the two versions include comparable material

and share analogous political perspectives. Each one (headline or article) was evaluated by at least five independent participants. In order to ensure unbiased assessments, participants were randomly assigned to content samples such that no individual saw both versions (original and neutralized) of the same item in the same session.

7.1.2 Measurements. The 7-point Likert scales were used for all assessments [6]. To determine the level of perceived prejudice, we asked:

To what extent do you see this headline (or narrative) as biased?

1—Completely impartial; 7—Completely biased.

In order to determine the article's or headline's ideology, we asked:

From a philosophical standpoint, what do you take out from this story or headline?

1—Extremely liberal; 4—Moderate; 7—Extremely conservative.

Our goal is to preserve political ideas by measuring content retention along three dimensions: *agreement on the preservation of the topic*, *political opinions*, and *semantic meaning*. According to Haim and Graefe [75], there were five factors that made up the readability scale: *being well-written*, *brief*, *thorough*, *cohesive*, and *clear*. Each item had to be evaluated on a scale from 1 (very low) to 7 (extremely high), as stated during the session. For this purpose, we asked if the polarity detector's recognized words were more polarizing than others in order to evaluate its performance. We used fabricated/randomly emphasized phrases to mitigate the priming effect and ensure that the responses were not influenced by the highlights.

7.2 Summary of Human Evaluation

7.2.1 Polarity Detector. In order to determine the relative polarity of the two groups of highlighted sentences, we asked participants to identify which ones were recognized by the polarity detector and which ones were emphasized at random by the researchers:

In comparison to the unselected phrases, do the highlighted statements show a higher level of political polarization?

In terms of political polarization, 68% of participants believed that the "real" highlighted words displayed more extreme views, while 32% of participants reached the same conclusion about the "fake" highlighted statements.

7.2.2 Content Preservation. We need to test how well our model preserves information, thus we asked participants three questions:

Question 1: Do you agree that the three aforementioned texts address the same subject matter?
Question 2: In your opinion, do all three of these writings present the same political viewpoints?
Question 3: In your opinion, are there any similarities in the meaning of the three passages described earlier?

We crafted the first question to be a moderate content-preserving evaluation. The second question was more challenging since it sought to evaluate the text in light of its implicit political viewpoints. The purpose of the last question was to evaluate semantic preservation. The results are shown in Table 14. The bulk of the audience felt that the texts' main points were similar across all three themes. When asked about the framework's performance in maintaining material for the body, more participants gave the impression that it was better than for the title. This may be attributed to

Table 14. The Proportion of Participants Who Agree That Our Neutralized Text Effectively Maintains the Content regarding Subject, Political Perspectives, and Semantic Similarities

	Content Preservation		
	Topic	**Political**	**Semantic**
Headline	92.65%	83.45%	92.56%
Text body	98.34%	91.67%	94.46%

Table 15. Summary of Human Evaluation Metrics and IRA

Evaluation Dimension	**Scale/Metric**	**Original Mean (SD)**	**Neutralized Mean (SD)**	**IRA**
Perceived Bias	7-point Likert (1 = Impartial, 7 = Biased)	5.82 (1.06)	2.91 (1.17)	Krippendorff's $\alpha = 0.71$
Ideological Orientation	7-point Likert (1 = Liberal, 7 = Conservative)	5.48 (1.13)	4.09 (0.96)	Krippendorff's $\alpha = 0.67$
Readability	Composite (5 traits, 1–7)	5.02 (0.84)	5.34 (0.79)	Krippendorff's $\alpha = 0.74$
Content Preservation: Topic	Binary (Agree/Disagree)	—	Headline: 92.65% Body: 98.34%	Fleiss' $\kappa = 0.73$
Content Preservation: Political	Binary (Agree/Disagree)	—	Headline: 83.45% Body: 91.67%	Fleiss' $\kappa = 0.69$
Content Preservation: Semantic	Binary (Agree/Disagree)	—	Headline: 92.56% Body: 94.46%	Fleiss' $\kappa = 0.72$

the greater length and contextual depth of full-length articles compared to headlines, making them more resilient to fluctuations.

7.2.3 ***Inter-Rater Agreement (IRA).*** We computed IRA across three core evaluation dimensions: perceived bias, ideological orientation, and readability. Specifically, we used Krippendorff's Alpha (α) to accommodate the ordinal nature of the Likert-scale responses and possible missing annotations. The resulting agreement scores were $\alpha = 0.71$ for perceived bias, $\alpha = 0.67$ for ideological classification, and $\alpha = 0.74$ for readability, indicating substantial agreement across the participants. For content preservation judgments, where binary agreement proportions were used (agree/disagree), we also computed Fleiss' Kappa, yielding $\kappa = 0.72$ for semantic similarity and $\kappa = 0.69$ for political viewpoint consistency, suggesting a high level of rater consensus. These metrics enhance the validity of our subjective evaluation framework.

It is evident from Table 15 that the neutralized versions significantly reduce perceived bias while maintaining ideological coherence and readability. Agreement among raters was substantial across all dimensions, with Krippendorff's Alpha exceeding 0.67 for subjective Likert ratings, and Fleiss' Kappa scores above 0.69 for binary content preservation judgments. These results reinforce the robustness of our framework's qualitative impact in comparison to human judgment.

8 Discussion

8.1 Theoretical Contribution

The current literature in the fields of computer science and social sciences is both improved by our investigations. The phenomenon known as confirmation bias causes people to subconsciously seek out news stories that support their current opinions, according to many studies on selective exposure [13].

According to research conducted by social scientists, news articles have a substantial impact on readers' political views [56]. Prior research develops an indicator of media bias to autonomously detect the most political expressions used by either Republicans or Democrats. As an example, according to Gentzkow [56], one study focuses on cataloging all phrases that appear in the Congressional Record and other articles. The frequency counting approach is significant but fails to account for varying circumstances. Our research tackles this problem via the implementation of a context-aware polarity detector.

In addition to identifying prejudice, our research offers an efficient method to mitigate readers' confirmation bias. In the past, researchers have looked into ways to give people different political news stories to read or by changing basic presentation techniques like using browser widgets to highlight agreeable content [139]. These highlighted presentation strategies, however, seem ineffective for persons resistant to challenges who detest confrontational messages [139]. Building upon existing research, our study investigates whether the presentation of news is neutralized or inverted by our transformer-based paradigm, which has the potential to diminish perceived bias, as shown by assessments indicating its efficacy in bias mitigation.

Our human assessment reveals that individuals' political ideologies influence their perceptions of news bias. Prior research on the negative impact of the media has shown that the bias in biased news coverage is seen differently by partisans based on the viewpoint they have toward politics [72]. Our findings are consistent with previous research. Moreover, our research introduces a novel methodology to investigate the hostile media impact by analyzing the differential reactions of partisans to shifts in news polarity (biased → unbiased). Our investigation reveals that our neutralizing tool may diminish the perceived bias disparities between two parties. Future research may investigate if the partisan divide may be diminished by presenting neutral news information to individuals with varying political affiliations.

8.2 Practical Implications

Both academics and practitioners in real-world environments have great potential to employ our methods. According to many scholars, media outlets will undoubtedly keep enabling the growth of online echo chambers due to cognitive bias and many social variables [170] unless something is done. Some media companies may try to appeal to politically biased audiences by excluding content that they see as offensive in order to maximize profits. As an alternate approach to reducing political bias and echo chambers, we provide our novel methodology for identifying polarity and then neutralizing polarized material.

8.2.1 Social Science Research. Social scientists have mostly focused on political polarization in the last several decades [16]. Our approach may be used by social scientists to assess different theories. Selective exposure academics propose that people often favor information that aligns with their attitudes, influenced by habitual news consumption and cognitive dissonance [100]. According to further research, people seek material that challenges their attitudes when they have the following three conditions: (a) a strong political connection and a substantial interest in politics; (b) the issue is very significant to them; and (c) the articles provide clearly comprehensible perspectives [100]. For example, Wojcieszak [218] predicted that previous studies often used real-life news stories as experimental stimuli. For example, to illustrate liberal and conservative viewpoints, respectively, [67] notes that several studies rely on national media like the *New York Times* and the *Wall Street Journal*. Despite the fact that many social scientists have attempted to identify stimuli that might influence the selection of news stories or headlines that represent different political ideologies, it remains challenging to eliminate potential biases stemming from disparities in writing quality or source credibility [109]. Our methodology can effectively reverse ideology

without affecting the semantic content or readability of news items, therefore it might be used to modify stimuli and reduce bias. After accounting for other factors, social scientists may look at the possibility that participant-aligned or neutral language can reduce the impact of selective exposure. By employing our method to generate comparable stimuli, other theories in the social sciences, such as testing for confirmation bias and the negative influence of the media, are also possible.

8.2.2 Practical Application. In addition to its use in academic research, our system has the potential to serve as a valuable tool for many practical applications. The prospective user demographics may encompass:

— *Content makers*, including journalists and editors, aim to engage broader audiences to ensure receptivity among individuals of varying ideological perspectives. Evidence from studies has demonstrated that news organizations and watchdog groups in the media, like *Ad Fontes Media*,[1] have made great strides in reducing media bias through various means, such as the creation of journalism ethics and standards. Our solution may significantly enhance the resources available to these content providers.
— *Content consumers* who pursue a wide array of political perspectives and knowledge, referred to as "diversity-seeking" persons. Current market tools either evaluate the political bias of media sources (e.g., *NoBias*[2] and *NewsGuard*[3]), or function as news aggregators that present a variety of news to readers (e.g., *ALLSIDES*[4]). Laboratory instruments [192] primarily concentrate on determining the ideal proportion of congenial and uncongenial information to maximally please readers. Our program does not only assign bias ratings to articles or curate a collection of impartial content for the customer. It enables news consumers to mitigate the bias of any selected item, so making the information more agreeable to their preferences.
— *Social media sites* use recommendation algorithms designed to enhance the visibility of material that users are predicted to interact with thereafter [170]. By altering the polarity/biasness of information to align more closely with users' preferences, social media platforms may provide identical material, although with varying polarity/biasness, to individuals across different political ideologies. This will enhance participation on social media sites while also mitigating filter bubbles.

8.3 Cross-Cultural Applicability and Domain Adaptability

CANDOR has been primarily evaluated using datasets such as ALLSIDES and BIGNEWS, both rooted in American political discourse. Political dichotomies such as liberal-conservative may not directly map to political typologies in Brazil (e.g., populist vs. institutionalist), or Europe (e.g., socialist vs. centrist). Bias cues vary not only in semantics and language, but also in regional media framing, editorial norms, and socio-political context. To tackle domain-specific shifts, future iterations of CANDOR may incorporate techniques from the style transfer and attribute-controlled generation literature. For example, the use of *semantic disentanglement* where ideological style is treated as a latent attribute, separate from content, enabling the system to decouple factual semantics from ideology markers. This is aligned with techniques proposed in [64, 113, 125, 131, 219, 221], and can be enhanced with adversarial objectives that minimize semantic drift while altering ideological tone. In low-resource transferability or multilingual regions, where labeled ideological corpora may be sparse, minimal supervision techniques may serve as viable alternatives and can be incorporated into the CANDOR pipeline [143]. Recent works [113, 122, 151] offer

[1] https://www.adfontesmedia.com.
[2] https://www.nobias.com.
[3] https://www.newsguardtech.com.
[4] www.allsides.com.

promising directions for future research in cross-cultural transfer. Moreover, *compliance-aware architectures* such as VLDBench [162] emphasize the importance of aligning automated rewriting and bias mitigation with local regulatory and platform-specific content norms. Inspired by their evaluation framework, CANDOR could optionally support *norm-conditioned decoding* or *policy-aligned generation* in future deployments.

In the context of multimodality, our VT-CT framework can further generalize across domains by allowing the contextual vectors to be anchored to new ideological axes (e.g., left-right in the UK, populist-elitist in Latin America, Left-Green-Right in German politics, or Communal-Secular in India). These anchor points can be mapped or projected using multilingual embeddings (e.g., XLM-R, mT5) and fine-tuned with local linguistic resources to retrain the fusion mechanism. Such adaptability ensures robustness even when visual cues (e.g., regional imagery, flag symbolism) differ culturally. In summary, while CANDOR offers state-of-the-art results in US-centric bias correction, its modular architecture—particularly the decoupled ideology encoder, conditional decoder, and context-aware fusion module—makes it amenable to adaptation across linguistic, cultural, and low-supervision domains. Future work may further explore integration of ideology disentanglement objectives, cross-lingual embedding projection, and policy-aligned generation to ensure equitable applicability across global news landscapes.

8.4 Computational Efficiency and Deployment Tradeoffs

The CANDOR framework demonstrates state-of-the-art performance involving multiple high-capacity components. These components include: (1) the BISLENS ideological encoder, (2) the DABDMN polarity detector [138], (3) the VT-CT encoder for multimodal context fusion, and (4) the CANDOR generative debiasing module using reinforcement learning. Each module is designed for a specific functional objective, and collectively they contribute to the overall computational complexity of the system.

The BIASLENS module, based on RoBERTa-large ($\sim$ 355M parameters), is fine-tuned using domain-adaptive pre-training with ideology-driven and story-driven triplet loss. Fine-tuning requires high GPU memory, $\sim$ 11 GB per instance on a V100 GPU. The DABDMN polarity detector, based on a RoBERTa backbone and trained with domain-specific objectives, has comparable complexity, but is optimized for discriminative inference, with reduced test-time latency. The VT-CT, introduced in Section 6.1 and elaborated in Table 6, plays a critical role in fusing multimodal representations, containing approximately 116M parameters, which is modest compared to BIASLENS and the generative modules. Nonetheless, its fusion layers add a linear overhead during inference, especially when processing high-resolution visual features alongside token embeddings. The CANDOR debiasing generator is trained via **Proximal Policy Optimization (PPO)** and evaluated across three transformer decoders: GPT-2 Medium (345M), FLAN-T5 Base, and LLaMA-2-7B. Among these, LLaMA-2 exhibited the most desirable results in terms of debiasing effectiveness and semantic preservation. However, this comes at a cost—policy rollouts and multireward evaluations extend average generation latency to $\sim$ 2.8 seconds per article with LLaMA-2. In contrast, GPT-2 reduces this to $\sim$ 1.6 seconds. On average, training time across the full pipeline (BIASLENS, VT-CT, polarity detection, and RL-based generation) is 4.7 hours per epoch on dual V100 GPUs.

To address this, we suggest several strategies. First, RL-based generation can be substituted with a supervised decoder trained via contrastive and content-alignment objectives, eliminating rollout overhead. Second, both BIASLENS and VT-CT can benefit from **Parameter-Efficient Fine-Tuning (PEFT)** techniques such as **Low-Rank Adaptation (LoRA)**, adapter modules, or low-rank compression. Third, the entire CANDOR pipeline can be compressed during distillation frameworks, transferring knowledge into lighter-weight models such as TinyLLaMA [234] or DistilRoBERTa [208]. Recent advances in PEFT methods offer promising avenues for mitigating the deployment

Table 16. Computational Breakdown and Efficiency—Performance Tradeoffs of CANDOR Modules

Component	Model Backbone	Params (M)	Role in Pipeline	Inference Cost	Tradeoff Options
BIASLENS	RoBERTa-large	355	Ideology-aware encoding (frozen during generation)	Low	LoRA, Adapters, Distillation
DABDMN	RoBERTa-base	125	Polarity classification using domain-adaptive pre-training	Low	Quantization, Multitask merge with BIASLENS
VT-CT (LMC + HFC)	Vision-Text Transformer	116	Multimodal fusion using layered and hierarchical context	Moderate	Low-rank compression, Pruning
CANDOR Generator	GPT-2/FLAN-T5/ LLaMA-2	345–7,000	Reinforcement learning-based neutral generation (PPO)	High	Supervised fine-tuning, LoRA, Model distillation

cost of LLMs without significantly compromising performance. Techniques such as LoRA [83], adapter layers [82], and prefix-tuning [114] enable selective fine-tuning of a small fraction of model parameters while keeping the majority of the backbone model frozen. For instance, LoRA injects learnable rank-decomposed matrices into each attention layer, allowing gradient updates to operate in a reduced parameter space. Applied to BIASLENS or the LLaMA-2 based CANDOR generator, LoRA can reduce fine-tuning memory requirements by over 65% and decrease training time by up to 50%, while preserving over 95% of the performance in domain-specific tasks. Adapter modules, which insert lightweight transformation blocks between existing layers, enable multitask adaptation without interference and are particularly useful when deploying CANDOR across ideologically diverse news domains. Incorporating such strategies would enhance the practical deployability of CANDOR on constrained hardware while preserving its fine-grained semantic and ideological sensitivity.

In summary, while CANDOR is computationally intensive by design, its architecture is inherently modular and flexible. This permits scalable deployment, allowing users to select tradeoffs between speed, accuracy, and explainability based on the specific application context. To offer a consolidated view of the computational footprint and potential optimization strategies across components of the CANDOR architecture, we present a summary in Table 16. This table outlines model backbone details, parameter size, functional roles, relative inference cost, and applicable tradeoff mechanisms. As seen, while modules like BIASLENS and the CANDOR RL generator are resource-intensive, various parameter-efficient strategies—such as LoRA, adapter-based tuning, and distillation—can be applied to enhance deployability without significantly sacrificing performance. This tabular overview complements our earlier discussion and reinforces CANDOR's modularity and adaptability in real-world deployment scenarios.

8.5 Findings

Our study delves into the topic of how language development might expose political bias in LLMs, along with a practical solution to alleviate such prejudice. Political bias in language models may exhibit the following characteristics:

Attributes Sensitivity: The bias of baseline language models fluctuates based on contextual factors (e.g., gender, place, subject), akin to other language integrity issues that are also context-dependent.

Imbalance among Ideologies: We highlight many intriguing discoveries about the disparity of political prejudice. It seems that left-leaning terms characterize the output of vanilla GPT-2. This might be attributed to the inherent liberal bias in the training data.

Both Lexical and Extra-Lexical: The political biases in produced phrases are not only lexical; instead, a contextual approach to debiasing is more successful than merely altering individual words.

It has been partially or completely shown that various types of bias have similar traits. One example is that gender bias in dialog systems depends on the utterance [119] and the persona [46]. There are disparities in salary estimation and crime rate prediction across different races and genders, according to [25] and [3]. Both human-created data and computer-generated text exhibit the same levels of political prejudice, according to our findings. Their findings showed that political bias in news articles is mainly shown by lexical variances. Using YouTube as a case study, Jiang et al. [92] found that a video's political orientation might impact its associated comments.

8.6 The Paradox of LLMs: A Double-Edged Sword

While the CANDOR framework utilizes LLM such as LLaMA-2 and GPT-2 to detect and neutralize ideological bias, it is essential to recognize the paradoxical role that LLMs often play in the broader context of media representation. These models enable nuanced identification of partisan language and facilitate the rewriting of news content into more balanced, less polarized forms. On the other hand, their generative flexibility can lead to unintended consequences—including semantic hallucinations, over-neutralization of meaningful ideological framing, or dilution of factual specificity.

Table 17 presents illustrative cases where the CANDOR model, despite its high level bias mitigation capability, produced outputs that either obscured important context, misrepresented the ideological spectrum, or softened legitimate criticism to the point of semantic vagueness. For example, in the case of climate policy (Row 2), a scientifically accepted reality is rephrased into a "debate," risking epistemic ambiguity. Similarly, systematic issues such as police brutality or racial injustice (Row 3) are reframed in vague terms like "diverse public responses," which may minimize the ethical gravity of the original statement.

This emergent behavior stems from the tension between informativeness and neutrality, a known limitation in auto-regressive generative modeling under value-sensitive constraints. As CANDOR aims for ideological balance, it may, in rare cases, undermine moral urgency or factual accuracy. These limitations highlight the importance of faithfulness-aware reward modeling, fact-grounded decoding strategies, and the incorporation of hallucination-detection filters in future iterations. We plan to address this duality through tighter coupling between truth-verification modules and bias-aware content planners, especially when applying CANDOR in high-stakes domains such as health, law or political communication.

9 Limitations and Potential Future Endeavors

Although the supposed occurrences are well captured by the bias measures that we look at, they undoubtedly include limits. Three types of political prejudice are the main emphasis of this article (*left-oriented*, *right-oriented*, and *center-oriented*) and would have to make major adjustments to include other types of bias with more class labels (for example, emotional bias, which is usually divided into nine categories [84]). Our second caveat is that we do not claim our results will work for any language since we only used English versions of modern language models in our analyses and trials. Nevertheless, our system may be expanded to include other languages by making the necessary adjustments, for instance, by using multilingual LMs like Multilingual-T5 [224]. Lastly, according to recent studies, people's views mirror their own political beliefs [136]. This intrinsic prejudice in individuals may possibly influence our human assessment. To mitigate this bias, we conducted our human assessments using ratings acquired from several annotators on the same set

Table 17. Examples Demonstrating the Paradoxical Behavior of LLMs during Bias Mitigation

Biased New (Left-oriented)	Biased News (Right-oriented)	Biased News (Center-oriented)	Neutral/Unbiased News
"The government's neglect of (immigrant children) at the border shows a [cruel, xenophobic agenda.]"	"(Illegal aliens) at the southern border are [exploiting weak immigration policies.]"	"(Detention conditions) at the border remain a [concern, with policy discussions ongoing.]"	"[Concerns] have been raised about the situation at the border, though official responses vary." ***(Loses detail on immigrant children; "official responses" is vague)***
"(Corporate greed) is to blame for [skyrocketing drug prices] that {hurt working families.}"	"(Regulations) have failed to control [Big Pharma], {harming middle-class Americans.}"	"[Drug prices have risen significantly] in recent years, with {multiple stakeholders involved.}"	"[Drug pricing] remains a (contentious issue), influenced by {market and policy dynamics.}" ***(Softens blame attribution to "market"; overlooks systemic causes)***
"(Climate change denial) fueled by {fossil fuel lobbies} continues to [threaten our planet.]"	"So-called (climate alarmists) push {costly agendas} without solid evidence."	"There is (political disagreement) over [climate change legislation.]"	"(Debate continues) around [climate change policy] and {environmental priorities.}" ***(Dilutes scientific consensus; frames it as debate)***
"(Police brutality) is a [systemic issue] rooted in {racial injustice.}"	"Critics exaggerate [isolated incidents] to (undermine the police.)"	"[Recent incidents] have sparked debate over (law enforcement practices.)"	"Concerns about (law enforcement practices) have prompted diverse public responses." ***(Avoids naming systemic racism; too vague to inform readers)***
"(Tax the rich) to fund [essential services like education and healthcare.]"	"Liberals push radical tax hikes that punish success and harm the economy."	"(Progressive taxation) proposals are gaining support amid (widening inequality.)"	"Proposals regarding tax reforms aim to [balance economic equity and growth.]" ***(Neutralizes to the point of policy ambiguity)***

While CANDOR aims to neutralize ideological language, it occasionally produces outputs that dilute factual clarity, obscure moral urgency, or introduce semantic vagueness—revealing the tension between neutrality and informativeness. The order of events is highlighted by red or (.), blue or [.], and magenta or {.} for better understanding.

of generated samples. Since our method cannot ensure the elimination of annotator bias, further effort is needed to enhance this section of the work.

9.1 Current Limitations and Conceptual Extensions

While our method does a good job of preserving text content, it does not guarantee that the underlying message will not be modified when the text bias/ideological polarity is changed. This is in addition to the aforementioned general shortcomings. The author's stylistic choices are sometimes intentionally selected to communicate a certain message. As our technology neutralized bias/polarity via changes in writing style, it may unintentionally impact the fundamental meaning. Observations on different impartial news produced are as follows:

Summarized News. Incapable of eliminating biases from intricate news, it ultimately generates a summary while retaining the prejudice (Table 18).

Chronological Error. LLM have difficulties in preserving the temporal or chronological sequence of events in news stories due to their absence of intrinsic awareness of real-world time-frames (Table 19). Although it may produce language that seems cohesive, it may lack a thorough comprehension of the chronological order of events.

Object Identification Error. It cannot discern the principal topic when many subjects are present in a given text (Table 20).

Table 18. Presented in a Biased Summary

Original Biased News	Generated Unbiased News
"In what activists are calling a clear case of (racial profiling, John Doe, a Black man, was detained yesterday by police in downtown Springfield). Eyewitness accounts suggest there was no apparent reason for the stop. (This incident adds to the growing list of grievances from the Black community, who see it as another example of the police's prejudiced behavior). Community leaders are decrying the systemic racism apparent in law enforcement, while police officials have yet to provide a statement."	"John Doe was stopped by police in Springfield, an event that some activists are labeling as [racial profiling. The Black community has cited this as part of ongoing issues with the police]. Officials have not commented on the incident."

(Left) The lines about "...racial profiling..." and the occurrence that contributes to it "This Incident Adds...," which are highlighted in blue or (.), are finally condensed into the news story about racial profiling, "...racial profiling. The Black...," that is created in an impartial manner (identified with orange or [.]).

Table 19. Chronological Mishap

Biased New (Left-oriented)	Biased News (Right-oriented)	Biased News (Center-oriented)	Neutral/Unbiased News
(Protests erupted in the city after the controversial decision was made by the government.) [The opposition party condemned the move], stating it was a violation of human rights. {This comes after weeks of tension between the government and activists.}	(The government took a decisive step to ensure law and order in the city, which has been plagued by unrest.) The opposition party's objections were expected, given their history of supportive disruptive elements. The move is seen as a necessary measure to protect citizens.	(Following a controversial decision, protests broke out in the city.) [The opposition party criticized the move], citing concerns about human rights violations. {The situation has escalated after weeks of tension between activists and the government.}	{After weeks of tension between activists and the government}, (protests broke out in the city following a controversial decision by the government.) [The opposition party criticized the move], citing concerns about human rights violations.

In all the three types of biased news, irrespective of their political orientation, the order in which the event took place followed a specific sequence like "Protest erupted in city after controversial decision by government... Opposition party stated it as violation of human rights... This happened after weeks of tension between government and activists." But the neutral/unbiased news shows somewhat a different chronology of events like "After weeks of tension between activists and government, protest broke out in the city... Opposition party criticized the move...." The order of events is highlighted by red or (.), blue or [.], and magenta or {.} for better understanding.

Previous studies have shown that LLMs are more resistant to generative bias, thus, future studies might focus on debiasing methods that are more complex and investigate the political bias in these models. In particular, the following studies might contribute to the field of bias research:

Explainability of Artificial Intelligence Systems. Elucidating the fundamental mechanics driving AI system judgments may enhance our comprehension of bias origins and provide insights for its mitigation. The widespread perception of modern deep learning models as enigmatic entities impedes their development of more effective methods for assessing and removing bias [117]. Enhanced transparency in AI systems would facilitate the identification of areas for future improvement and optimization.

Behavioral Analysis in Sociology. The majority of recent studies on AI bias have focused on AI system-level causes and consequences, largely ignoring the role of human and sociological variables. Many recommendation systems use popularity bias to provide information aligned

Table 20. Object Identification Mishap

Biased New (Left-oriented)	Biased News (Right-oriented)	Biased News (Center-oriented)	Neutral/Unbiased News
The (government's decision to raise taxes on the wealthy elite and corporations has sparked outrage.) [Critics argue that the move will hurt economic growth and lead to job losses.] However, {supporters believe it is necessary to fund vital social programs and reduce income inequality.}	The (government's tax hike on the wealthy and corporations is a necessary step to ensure fiscal responsibility.) [Critics of the move fail to see that it will lead to greater economic stability and prosperity for all.] {It's time to put the country's financial health first.}	The (government's decision to raise taxes has drawn mixed reactions.) [Critics fear it could harm economic growth], while {supporters argue it's essential for funding social programs and reducing income inequality.}	(Reactions to the government's decision to raise taxes are varied.) [Supporters argue it could impact economic growth and lead to job losses], while {others believe it's necessary for funding social programs and reducing income inequality.}

The neutral/unbiased news changed the narration completely. The biased news states Government's decision to raise tax has sparked rage… Critics argue the move will hurt economic growth… Supporters believe it is necessary to fund social programs…. On the other hand, the neutral/unbiased news narrates the news as "Reactions of the government's decision to raise tax varied… Supporters argued it could impact economic growth… Others believe it's necessary for funding social programs…." The order of events is highlighted by red or (.), blue or [.], and magenta or {.} for better understanding.

with audience interests [32]. Research indicates that individuals have a preference for consuming news that aligns with their political biases and seek out perspectives that resonate with their own predispositions. We suggest that improved conceptualization and mitigation techniques might be achieved by including social variables.

Efficiency vs. Fairness. Research in machine learning fairness has validated the tradeoff between algorithmic efficiency—where neglecting fairness permits unconstrained optimization—and fairness [41]. We address political bias by implementing additional computations on biased terms or using a classifier, which therefore increases the delay in language model construction. Future research may provide novel debiasing approaches that do not significantly compromise performance.

Varied Examination. In order to make our dataset more representative of society and its political spectrum, future studies may expand our news story inclusion to include items from autonomous, non-profit, or new startup media corporations. To further understand how different subjects (such abortion and gun control) impact partisans' perceptions of changed news narratives, future studies should more fairly distribute topics, considering that partisans may have different viewpoints on many issues. One potential future direct use of our study is to find the sweet spot for the number of neutralized original sentences that provide the most delightful reading experience.

9.2 Cross-Domain Applications of Bias Mitigation Frameworks

9.2.1 Extending CANDOR for Network Immunization. Misinformation and political bias increasingly propagate through social platforms. Network immunization strategies for harmful content mitigation may be extended by exploring CANDOR's multimodal bias detection capability into graph-based immunization settings by ranking influential users, communities, or content cascades based on their ideological shift. Dynamic node immunization methods [228], adopt real-time heuristics and gain functions to iteratively protect influential nodes. Sparse Shield [149] introduced rank suppression for harmful speech using sparse graph eigenvalue minimization. Similarly, Truică et al. [201] proposed MCWDST, which uses minimum-cost spanning tree algorithms to suppress fake news diffusion based on diffusion latency, speed, and breadth. CANDOR's RL-based debiasing

engine could be adapted to selectively neutralize biased content from high-risk nodes, informed by real-time diffusion and influence graphs. From a community-based perspective, models like CONTAIN [10] and CHA [158] propose identifying vulnerable communities via structural metrics like boundary node trust, community believability, and exposure susceptibility. Integrating such vulnerability scores with CANDOR would enable localized immunization strategies, protecting communities from ideological contagion. By embedding the mechanism reported in [223] into CANDOR, future research can aim to proactively block the emergence of ideological polarization by using bias-informed immunization planning.

9.2.2 Extending CANDOR for Harmful Content Detection. The real-time detection of misinformation and harmful content, particularly in crisis-driven or disaster-related contexts—has gained traction through distributed systems like ContCommRTD, which use temporal, textual, and community cues for content-level moderation [11]. A future extension of CANDOR could similarly integrate dynamic community modeling and temporal embeddings to handle event-centric harmful content. The work in [9] discussed detection of coordinated malicious communities.

With the rise of immersive platforms, a critical gap exists in handling harmful content in virtual societies and metaverse ecosystems [217]. CANDOR's multimodal framework can evolve to incorporate audio-visual misinformation, hybrid attention mechanisms, and VR-based trust scoring, supporting content moderation beyond traditional text. Additionally, robust and explainable detection mechanisms such as those proposed in PoliGraph offer fault-tolerant and adversarially secure frameworks using graph neural networks and Byzantine fault-tolerant nodes [174]. CANDOR could incorporate such resilience models to ensure stable debiasing even under hostile misinformation environments.

Recent survey work [12] emphasizes the need to bridge the gap between academic efforts and platform-level demands, advocating for scalable, multilingual, explainable, and culturally sensitive solutions. CANDOR's RL-based rewriting strategy is well suited for such extensions, particularly when supported by language-agnostic transformers like XLM-R [40] or mT5 [224]. Moreover, STOPHC, a combined architecture for detection and immunization, demonstrates the utility of deep ensemble classifiers and immunization algorithms, like NetShield and DAVA to block harmful information flow across user graphs [204]. A modular extension of CANDOR could enable joint bias and harm detection, integrating text classifiers and influence-aware immunization.

9.2.3 Extending CANDOR for Virality Detection. The work [203] introduces a temporal attention-driven neural architecture that accurately predicts virality using early audience reactions, sentiment, and topic sensitivity. An extension of CANDOR would integrate such early-stage predictors into its multimodal pipeline, allowing it to proactively intervene in the propagation of highly biased or polarizing content. Furthermore, CANDOR could implement virality-aware debiasing thresholds, triggering rewriting or content warnings only when ideological bias coincides with projected audience amplification. This dual-scoring framework can support both moderation prioritization and news feed re-ranking, aligning well with platform deployment needs. Future work may also explore fusion architectures combining early engagement signals, source credibility, stance detection, and network amplification patterns, extending CANDOR into a predictive risk engine for ideological virality.

9.3 Positioning CANDOR among Emerging Bias Mitigation Paradigms

Recent advancements in responsible LLM design have opened up alternative pathways for bias mitigation, emphasizing computational efficiency, data efficiency, and interpretability [159]. CANDOR's positioning can be appreciated by contrasting it with recent lightweight and annotation-scarce

Table 21. Comparison of CANDOR with Emerging Bias Mitigation Paradigms

Method Class	Resource Intensity	Interpretability	Annotation Need
CANDOR (Ours)	High (multimodal, fine-tuned)	Medium–High (attention + polarity maps)	Medium–High (BIGNEWSALIGN, ALLSIDES)
LoRA-Based Finetuning [83]	Low–Medium	Low	Low
Prompt-Based Debiasing [102]	Low	Medium	Low
Constrained Decoding [178]	Medium	High	Medium
Masked Rationale Learning [99]	Medium	High	Medium

paradigms. Several studies have proposed that bias mitigation is a *lightweight fine-tuning* mechanism such as LoRA [83], which achieves task alignment while preserving base model semantics. In contrast, CANDOR's fine-tuning incorporates ideology and story-driven objectives in the BIASLENS module, outperforming LoRA-like methods in nuanced political datasets though heavier in terms of computational overhead.

Another axis of contrast lies in *annotation scarcity*. Prompt-based debiasing methods [102, 159] and in-context learning demonstrate potential for rapid generalization using minimal supervision. CANDOR, by contrast, leverages *rich story-aligned datasets* (e.g., BIGNEWSALIGN) and curated multimodal pairs to train context-aware debiasing objectives. While this offers superior semantic preservation and content fidelity, it raises challenges in scaling to low-resource domains. Hence, semi-supervised or pseudo-labeling variants of BIASLENS can serve as future directions. Finally, interpretable debiasing techniques, such as constrained decoding [178], rule-augmented generation [171], or token-level rationale attribution [99], prioritize human-aligned transparency. CANDOR complements these by triangulating semantic, lexical, and concept-based polarity metrics (Section 6.4) along with visualizing ideological attribution maps (Section 6.5). In summary, while CANDOR pursues a comprehensive, multimodal strategy for political bias mitigation, it aligns with and complements recent minimalistic paradigms. Table 21 outlines a high-level comparison across three axes: resource requirements, interpretability, and domain generalization, positioning CANDOR within the broader spectrum of emerging bias mitigation frameworks.

10 Conclusion

This research adds full-length articles to the current political polarity headline-flipping task and introduces an innovative architecture designed to address two principal challenges in this assignment: (1) Identifying politically polarized material within an extensive article; (2) Mitigating the polarized/biased text while preserving content integrity and fluency. Using the Transformer design, we proposed a polarity detector and a polarity debiaser. For this purpose, we conducted evaluations of our methods using both automatic and manual tools. In a number of assessment tasks that allow for direct comparisons, our approach surpasses the performance of the most recent state-of-the-art model. Human assessments validate that, from the perspective of impartial observers, we have developed a method that successfully neutralizes headlines and full-length articles without compromising content integrity or overall fluency. Although assessments indicate that our model is operating well, considerable opportunities for enhancement remain. The process of altering the neutralizing polarity of articles is hindered by an absence of defined definitions and, more critically, a deficiency of standardized benchmark datasets. This effort attempts to tackle this problem by creating a large annotated dataset and clearly outlining the aim and its subtasks. Our goal is for our paradigm to reduce political polarization and assist those working in the field of social science in dismantling echo chambers. A direct continuation of this study includes enrolling trial participants as communicators and conducting randomized controlled trials to evaluate its usefulness in addressing filter bubbles. However, this is outside the scope of this publication.

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